AWS Solutions Architect

Professional (SAP-C02) — Study Notes

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Study strategy for SAP

SAP builds directly on SAA. These notes focus on **depth and breadth beyond the associate level**:

- SAP blocks in the shared service files are your primary SAP study material.
- This document adds domain-level context that is *SAP-only* (multi-account, migration, governance).
- If a concept is tagged SAA+SAP in the shared files, treat it as assumed knowledge here.

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Part I

Exam Overview

1 SAP-C02 at a Glance

[key] Key facts

- **Questions:** 65 scored + 10 unscored = 75 total
- **Duration:** 180 minutes
- **Passing score:** 750 / 1000
- **Format:** Multiple choice + Multiple response (often 3-of-5 or 2-of-5)
- **Prerequisite (recommended):** SAA-C03 or equivalent experience

1.1 Domain Weightings

SAP-C02 content domains

Domain	Name	Weight
1	Design Solutions for Organizational Complexity	26%
2	Design for New Solutions	29%
3	Continuous Improvement for Existing Solutions	25%
4	Accelerate Workload Migration and Modernization	20%

How SAP differs from SAA:

- Scenarios are longer and involve *trade-offs* between multiple valid options
- Heavy focus on **organisational complexity**: multi-account, hybrid, enterprise
- Migration and modernisation at scale
- Cost and operational *governance*, not just individual architecture

2 Domain Map Overview

SAP questions are scenario-heavy. The right answer often depends on *organisational constraints* (existing on-prem, multi-account, compliance). Always read the full scenario before eliminating options.

2.1 Domain 1 — Design Solutions for Organizational Complexity (26%)

Domain 1 — Design Solutions for Organizational Complexity

Key services: AWS Organizations, Control Tower, Service Control Policies (SCPs), IAM Identity Center, Resource Access Manager (RAM), Service Catalog, Config Aggregator, Security Hub (multi-account), CloudTrail (org trail), Cost Allocation Tags.
Core tasks:

- Architect network connectivity strategies for complex environments (Transit Gateway, Direct Connect, VPN hub-and-spoke)
- Prescribe security controls for multi-account environments (SCPs, permission boundaries, IAM Identity Center)
- Design multi-account management strategies (AWS Organizations, Landing Zone)
- Determine cost optimisation opportunities across an organization

Watch for keywords: "hundreds of accounts," "centrally enforce," "prevent member accounts from," "shared services VPC," "org-wide CloudTrail."

2.2 Domain 2 — Design for New Solutions (29%)

Domain 2 — Design for New Solutions

Key services: All SAA services + EKS, AppSync, EventBridge Pipes, Step Functions, MWAA, Kinesis Data Streams/Firehose, OpenSearch, Lake Formation, Glue, AppFlow, API Gateway (private/edge/regional), ElastiCache Global Datastore.
Core tasks:

- Design a deployment strategy to meet business requirements (canary, blue/green)
- Design a solution to ensure business continuity (active-active, global)
- Determine security controls based on requirements
- Design a reliable and resilient architecture
- Design a solution to meet performance objectives
- Determine a cost-optimised design

Watch for keywords: "zero-downtime deployment," "near-zero RPO," "event-driven," "microservices," "global users," "high throughput ingestion."

2.3 Domain 3 — Continuous Improvement for Existing Solutions (25%)

Domain 3 — Continuous Improvement for Existing Solutions

Key services: Trusted Advisor, Compute Optimizer, Cost Explorer, Systems Manager, Config Rules (auto-remediation), CloudWatch (alarms, dashboards, anomaly detection), X-Ray, CloudTrail, DevOps Guru, Resilience Hub.
Core tasks:

- Determine a strategy to improve overall operational excellence
- Determine a strategy to improve security (automated remediation, detective controls)
- Determine a strategy to improve reliability
- Determine a strategy to improve performance
- Determine a strategy to improve cost optimisation

Watch for keywords: "improve existing," "reduce operational overhead," "automate remediation," "identify underutilised," "reduce costs on current architecture."

2.4 Domain 4 — Accelerate Workload Migration and Modernization (20%)

Domain 4 — Accelerate Workload Migration and Modernization

Key services: Application Migration Service (MGN), Database Migration Service (DMS), Schema Conversion Tool (SCT), Migration Hub, VM Import/Export, DataSync, Snow Family (Snowball Edge, Snowcone, Snowmobile), Transfer Family, Mainframe Modernization.
The 7 Rs of Migration:

1. **Retire** — decommission, no longer needed
2. **Retain** — keep on-prem (compliance, not ready)
3. **Rehost** (Lift & Shift) — move as-is to EC2/MGN
4. **Relocate** — move hypervisor to VMware Cloud on AWS
5. **Replatform** (Lift & Reshape) — minor optimisation (EC2 → RDS)
6. **Repurchase** — move to SaaS (e.g. CRM → Salesforce)
7. **Refactor/Re-architect** — redesign for cloud-native (Lambda, containers)

Watch for keywords: "migrate with minimal downtime," "large dataset transfer," "heterogeneous database migration," "containerise existing app," "modernise mainframe."

[!] Migration exam tip

When the question mentions **terabytes of data** and a **slow/expensive link**, the answer is almost always a **Snow Family** device, not DataSync or Direct Connect. DataSync is best for ongoing, incremental sync over an existing network connection.

Part II

AWS Foundations (SAP depth)

3 AWS Global Infrastructure

SAA *Regions, AZs, Edge locations*

[key] Key facts

- **Region** — geographically isolated area; contains 2+ AZs (typically 3). Choose region based on: latency, data residency, service availability, pricing.
- **Availability Zone (AZ)** — one or more discrete data centres with independent power, cooling, networking. AZs in a region are interconnected via high-bandwidth, low-latency private fibre.
- **Edge Location / Point of Presence (PoP)** — used by CloudFront (CDN) and Route 53 for caching and DNS resolution closer to end users. More locations than Regions.
- **Local Zone** — extension of an AWS Region to metropolitan areas; for ultra-low latency workloads (gaming, media, real-time).
- **Wavelength Zone** — AWS infrastructure embedded in 5G carrier networks for single-digit millisecond latency to mobile devices.
- **Outposts** — AWS-managed rack/server hardware in your own data centre; extends AWS APIs on-premises.

3.1 Resilience by Design

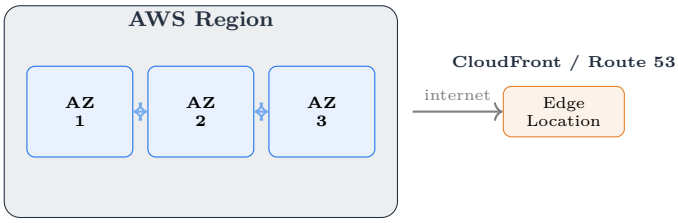
Design for failures at every level:

- **AZ-resilient** — deploy across ≥ 2 AZs; use Multi-AZ RDS, ALB, ASG spread.
- **Region-resilient** — cross-region replication (S3, Aurora Global, DynamoDB Global Tables).
- **Global** — Route 53 latency/health-check routing, CloudFront origin failover.

[!] Exam tip

Questions about "highest availability" or "survive an AZ failure" always point to **Multi-AZ deployments**. Questions about "survive a region failure" require cross-region architectures — Aurora Global Database, S3 Cross-Region Replication, or Route 53 failover.

3.2 Global Infrastructure Diagram



[*] SAP depth

Multi-Region patterns (SAP focus):

- **Active–Active**: traffic served from multiple regions simultaneously (Route 53 latency-based routing). Requires data synchronisation (DynamoDB Global Tables, Aurora Global write forwarding).
- **Active–Passive**: primary region handles all traffic; secondary region is on standby (warm standby or pilot light). Promotes on failover.
- **Data residency**: use AWS Organizations SCPs to `Deny` API calls outside approved regions — enforces sovereignty requirements.
- **AWS GovCloud**: isolated US regions for regulated workloads (FedRAMP, ITAR). Separate account required; accessed via specific endpoints.

4 AWS Well-Architected Framework

SAA+SAP *The six pillars that guide every SAA and SAP architecture question.*

The Well-Architected Framework gives you a consistent way to evaluate architectures and implement designs that scale over time. **Every exam domain maps back to at least one pillar.**

4.1 The Six Pillars

[key] Key facts

1. **Operational Excellence** — run, monitor, and improve systems. Focus on automating operations and responding to events.
2. **Security** — protect data, systems, and assets. Includes IAM, detective controls, infrastructure protection, and incident response.
3. **Reliability** — recover from failures and meet demand dynamically. Covers HA, DR, and fault tolerance.
4. **Performance Efficiency** — use resources efficiently as demand changes; choose the right resource type and size.
5. **Cost Optimization** — avoid unnecessary costs; match supply to demand; use the right pricing model.
6. **Sustainability** — minimise environmental impact of cloud workloads. (Newer pillar — appears more on SAP.)

[!] Exam tip

When an exam question asks you to choose between two architecturally valid options, ask yourself: *which pillar is the question testing?* Keywords like "least operational overhead" → Cost/Operational Excellence; "most fault-tolerant" → Reliability; "most secure" → Security.

4.2 Design Principles (per pillar)

4.2.1 Operational Excellence

- Perform operations as code (Infrastructure as Code)
- Make frequent, small, reversible changes
- Anticipate failure (game days, chaos engineering)
- Learn from all operational events and failures

4.2.2 Security

- Implement a strong identity foundation (least privilege)
- Enable traceability (CloudTrail, Config, GuardDuty)
- Apply security at all layers
- Protect data in transit and at rest (KMS, TLS)
- Keep people away from data (automate where possible)

4.2.3 Reliability

- Test recovery procedures
- Automatically recover from failure
- Scale horizontally to increase aggregate system availability
- Stop guessing capacity — use Auto Scaling
- Manage change using automation

4.2.4 Performance Efficiency

- Democratise advanced technologies (use managed services)
- Go global in minutes (multi-region)
- Use serverless architectures
- Experiment more often
- Consider mechanical sympathy (use the right tool for the job)

4.2.5 Cost Optimization

- Implement cloud financial management (FinOps)
- Adopt a consumption model (pay for what you use)
- Measure overall efficiency (CloudWatch metrics per $)
- Stop spending money on undifferentiated heavy lifting
- Analyse and attribute expenditure (tagging, Cost Explorer)

4.2.6 Sustainability

- Understand your impact (carbon footprint tool)
- Maximise utilisation (right-size, use Graviton)
- Adopt managed services (AWS manages hardware lifecycle)

[*] SAP depth

At the SAP level, questions tie the framework to **organisational strategy**: how do you enforce pillars across multiple accounts? Key tools:

- **AWS Well-Architected Tool** — workload reviews, milestone tracking
- **AWS Trusted Advisor** — automated best-practice checks (5 categories map to pillars)
- **AWS Control Tower** — landing zone that enforces guardrails (SCPs) at scale
- **AWS Service Catalog** — approved product portfolios → governance at scale

[!] SAP exam tip: "most operationally efficient"

SAP questions often say "with the **least operational overhead** going forward." This almost always points to a **managed/serverless** service over a self-managed EC2 solution. E.g. Aurora Serverless over RDS on EC2, or Fargate over EC2-backed ECS.

Part III

Compute (SAP depth)

5 Amazon EC2

SAA+SAP *Virtual servers in the cloud. The most fundamental AWS compute service.*

5.1 What is EC2?

EC2 provides resizable compute capacity in the cloud. You rent virtual machines (called **instances**) rather than buying physical hardware. You choose the OS, CPU, RAM,

storage, and network configuration.

[key] Key facts

- Instances run inside a **VPC** in a specific **Availability Zone**
- Billing: per-second (Linux) or per-hour (Windows) — minimum 60 seconds
- An **AMI** (Amazon Machine Image) defines the OS + pre-installed software
- **User data** scripts run once at first boot (bootstrap automation)
- **Instance metadata** is accessible at `169.254.169.254`

5.2 Instance Types

Instance types follow a naming pattern: `<family><generation>.<size>` e.g. `m6i.large`, `c5.xlarge`, `r6g.2xlarge`.

Instance family cheat-sheet

Family	Use case	Examples
General Purpose	Balanced CPU/RAM/network	`m6i, t3, t4g`
Compute Optimised	High CPU ratio	`c6i, c7g`
Memory Optimised	Large in-memory workloads	`r6i, x2idn, u-*`
Storage Optimised	High sequential I/O	`i3, i4i, d3`
Accelerated	GPU / ML inference	`p4, g5, inf2`

[!] Exam tip

T-family (burstable): T3/T4g instances accumulate CPU credits when idle and spend them during spikes. Use `unlimited` mode to avoid throttling at extra cost. Exam favourite for variable-load web apps with a low baseline.

5.3 Purchasing Options

Pricing models

Model	Best for	Discount vs On-Demand
On-Demand	Short-term, unpredictable	—
Reserved (1 or 3yr)	Steady-state, predictable load	up to 72%
Savings Plans	Flexible RI alternative	up to 66%
Spot	Fault-tolerant, flexible workloads	up to 90%
Dedicated Host	License compliance (BYOL)	varies
Dedicated Instance	Single-tenant hardware (no BYOL)	varies
Capacity Reservation	Reserve AZ capacity (no discount)	—

[!] Exam tip

Spot Instance interruption = 2-minute warning via instance metadata and EventBridge. Architect around this with *Spot + On-Demand mixed* Auto Scaling groups. Keywords: "lowest cost," "fault-tolerant," "batch jobs."

[!!] Gotcha

Reserved Instances apply to a **specific AZ** (Zonal RI → capacity reserved) or **any AZ in a region** (Regional RI → no capacity reservation but flexible). Exam will test whether you need the capacity guarantee.

5.4 Storage for EC2

[key] Key facts

- **EBS (Elastic Block Store)** — network-attached, persists after instance stop/terminate (if configured). One EBS ↔ one AZ.
- **Instance Store** — physically attached NVMe; *ephemeral* (lost on stop/terminate). Fastest possible IOPS. Good for caches/scratch.
- **EFS** — shared NFS mount across multiple instances/AZs.
- **FSx for Windows** — SMB share for Windows workloads.

5.4.1 EBS Volume Types

EBS types

Type	Use case	Max IOPS	Max throughput
gp3 (SSD)	General purpose boot/dev	16,000	1,000 MB/s
gp2 (SSD)	Legacy general purpose	16,000	250 MB/s
io2 Block Expr	Critical DBs (sub-ms)	256,000	4,000 MB/s
io1 (SSD)	I/O intensive DBs	64,000	1,000 MB/s
st1 (HDD)	Throughput-heavy big data	500	500 MB/s
sc1 (HDD)	Cold, infrequent access	250	250 MB/s

[!] Exam tip

gp3 vs gp2: gp3 lets you provision IOPS and throughput *independently*. gp2 ties IOPS to size (3 IOPS/GB, max 16,000). Default to gp3 for new workloads.

5.5 Networking

[key] Key facts

- Each instance gets a **primary private IP** in its subnet (persists)
- **Elastic IP (EIP)** — static public IP you own; reassignable. **Cost note:** Charged when *not* attached to a running instance.
- **ENI (Elastic Network Interface)** — virtual NIC; you can attach multiple ENIs to an instance for network separation.
- **Enhanced Networking** — SR-IOV for higher throughput, lower latency. Enabled via ENA driver (most modern instances) or Intel 82599 VF.
- **Placement Groups**:
  - *Cluster* — same rack, lowest latency, 10 Gbps between instances.
  - *Spread* — different hardware, max 7 instances/AZ.
  - *Partition* — racks isolated per partition; for Hadoop/Kafka/HDFS.

5.6 EC2 Auto Scaling

Auto Scaling maintains the *desired capacity*, replacing unhealthy instances and scaling in/out based on policies.

[key] Key facts

- **Launch Template** (preferred) or Launch Configuration (legacy)
- **Scaling policies**:
  - *Target Tracking* — keep metric at a value (e.g. CPU = 50%). Simplest; AWS manages scale-in/out automatically.
  - *Step Scaling* — scale by different amounts at different thresholds.
  - *Scheduled* — predictable load patterns (e.g. weekday 9am).
  - *Predictive* — ML-based, pre-warms capacity.

- **Cooldown period** — default 300 s; prevents thrashing.
- **Warm-up period** — time before new instance counts toward metrics.
- **Lifecycle hooks** — pause instance at launch/terminate for custom logic.

[!] Exam tip

"Scale fast, scale slow": Target tracking with a low scale-out cooldown handles sudden spikes. Add a longer scale-in cooldown to avoid premature termination. Pair with an *ALB* for HTTP(S) and *NLB* for TCP.

5.7 Security

[key] Key facts

- **Security Groups** — stateful firewall at the instance/ENI level. Default: allow all outbound, deny all inbound.
- **IAM Instance Profile** — grants the instance an IAM role. *Never* store credentials on an instance; use roles instead.
- **Key Pairs** — RSA key for SSH (Linux) / RDP password decrypt (Windows). AWS stores the public key only.
- **EC2 Instance Connect** — browser-based SSH without long-lived keys.
- **SSM Session Manager** — shell access without SSH, no open port 22 needed. Preferred for secure environments.

[i] SAA depth

For the SAA exam, understand the **Security Group vs NACL** distinction: Security Groups are stateful (return traffic auto-allowed), NACLs are stateless (return traffic needs explicit allow). NACLs apply at the subnet level.

[*] SAP depth

SAP scenarios involve **migrating large EC2 fleets**:

- **AWS Migration Hub** — track migration progress centrally.
- **AWS Application Migration Service (MGN)** — lift-and-shift at scale; replicates disks continuously, minimal downtime cutover.
- **VM Import/Export** — one-off migrations from VMware/Hyper-V.
- Right-sizing: use **AWS Compute Optimizer** (ML recommendations) or **Trusted Advisor** for over-provisioned instances.
- **Mixed-instance policies** in ASG: combine Spot + On-Demand with multiple instance families/sizes for cost and availability.

[!!] Gotcha

Stopping vs Terminating: a **stopped** instance retains its EBS root volume; a **terminated** instance deletes it (unless `DeleteOnTermination=false`). Hibernate

preserves RAM to EBS — instance must be EBS-backed, encryption enabled.

6 AWS Lambda

SAA+SAP *Serverless compute — event-driven, pay-per-invocation*

[key] Key facts

- Runs code in response to events; AWS manages servers, scaling, patching.
- **Supported runtimes**: Node.js, Python, Java, Go, Ruby, .NET, custom (container image).
- **Max execution time**: 15 minutes (900 s) per invocation.
- **Memory**: 128 MB – 10 240 MB (CPU scales proportionally).
- **Deployment package**: ZIP ≤50 MB direct / 250 MB unzipped; up to 10 GB via container image.
- **Ephemeral storage** (`/tmp`): 512 MB – 10 240 MB.
- **Concurrency**: **Default:** 1 000 concurrent executions per region (soft limit).
- **Pricing**: $0.20 per 1 M requests + duration (GB-s); generous free tier.

6.1 Invocation Models

Lambda invocation models

Model	How it works	Common triggers
Synchronous	Caller waits; errors returned to caller	API Gateway, ALB, Cognito, CloudFront@Edge
Asynchronous	Lambda queues event; retries ≤2×; DLQ for failures	S3 events, SNS, EventBridge, CloudWatch Events
Event Source Mapping	Lambda polls source; processes in batches	SQS, Kinesis, DynamoDB Streams, MSK

[!] Exam tip

SQS + Lambda (event source mapping): Lambda polls SQS. Messages are deleted only after successful processing. On failure they become visible again; after max retries they go to a DLQ. This is the canonical *decouple-and-process* pattern for SAA questions.

6.2 Networking — Lambda in a VPC

By default Lambda runs in an AWS-managed VPC with internet access. To connect to *private* resources (RDS, ElastiCache, private APIs):

- Configure Lambda with a VPC, subnet(s), and Security Group.

- Lambda injects **Elastic Network Interfaces (ENIs)** in the specified subnets.
- Private subnet ⇒ needs **NAT Gateway** for internet; or use **VPC Endpoints** for AWS services (S3, DynamoDB, etc.).
- Cold start latency increases slightly; mitigated with **Provisioned Concurrency**.

[!!] Gotcha

Lambda in a VPC *loses internet access* unless a NAT Gateway is present. This is the #1 Lambda VPC trap: Lambda can reach private RDS but cannot call external APIs or reach S3/DynamoDB without a NAT Gateway or VPC Endpoint.

6.3 Concurrency & Scaling

- **Reserved concurrency**: hard cap on a function's concurrent executions. Guarantees capacity; prevents one function from starving others.
- **Provisioned concurrency**: pre-initialises execution environments; eliminates cold-start latency — essential for latency-sensitive APIs.
- **Burst concurrency**: Lambda can scale from 0 to thousands of instances in seconds (pool-based initial burst, then linear growth).

6.4 Common Architecture Patterns

- **REST API**: API Gateway → Lambda → DynamoDB / RDS (via RDS Proxy)
- **File processing**: S3 `PutObject` event → Lambda (image resize, ETL)
- **Stream processing**: Kinesis / DynamoDB Streams → Lambda (event source mapping)
- **Fan-out**: SNS → Lambda (asynchronous; multiple subscribers possible)
- **Scheduled tasks**: EventBridge scheduled rule → Lambda (cron replacement)
- **Auth at edge**: Lambda@Edge or CloudFront Function for JWT validation, A/B testing

[i] SAA depth

Lambda vs EC2 — when to choose:

- **Lambda**: short-lived tasks (<15 min), event-driven or spiky traffic, no server management desired, per-request billing is cost-effective.
- **EC2**: long-running processes, custom OS / runtime needs, GPU compute, sustained high-throughput where per-invocation pricing becomes expensive.
- Lambda cannot keep persistent database connections across invocations — use **RDS Proxy** to pool connections in Lambda-to-RDS architectures.

[*] SAP depth

Lambda advanced topics (SAP):

- **Lambda Destinations**: on success/failure route results to SQS, SNS, EventBridge, or another Lambda. Richer than DLQ; preserves invocation metadata.
- **Lambda Extensions**: attach observability agents (Datadog, Splunk) as layers without modifying function code.
- **Container Images**: package as Docker image (≤10 GB from ECR); same Lambda runtime model — good for large ML models or custom runtimes.
- **Step Functions + Lambda**: orchestrate multi-step workflows; Lambda handles individual steps; Step Functions manages retries, branching, and state.
- **ALB vs API Gateway**: ALB cheaper for simple HTTP routing to Lambda; API Gateway adds auth, throttling, usage plans, WebSocket, request/response transformation — justified for public-facing APIs.
- **Multi-region**: identical Lambda + API Gateway stacks per region; Route 53 latency-based or health-check failover routing.
- **Lambda SnapStart (Java)**: initialises function on version publish, snapshots memory/disk state; near-zero cold start for Java runtimes.

[!] Exam tip

Key limits for exam: 15-min max timeout, 10 GB RAM, 10 GB /tmp, 10 GB container image, 1 000 default concurrent executions per region. If a question mentions a "15-minute processing job" running on Lambda — that is the maximum; anything longer must use EC2, ECS, or Batch.

7 EC2 Auto Scaling

SAA+SAP *Automatically adjust EC2 fleet size based on demand*

[key] Key facts

- **Auto Scaling Group (ASG)**: logical group of EC2 instances; managed as a fleet.
- Defines: **Minimum**, **Desired**, and **Maximum** capacity.
- **Launch Template** (preferred) or Launch Configuration: specifies AMI, instance type, key pair, SG, IAM role, user data for new instances.
- Integrates with ELB: new instances are automatically registered with the target group; unhealthy instances are replaced.
- **Health checks**: EC2 status check (default) or ELB health check (more precise — app-level).
- **AZ balance**: ASG distributes instances evenly across specified AZs.

7.1 Scaling Policies

Auto Scaling policy types

Policy	How it works	Use case
Target Tracking	Maintains a target value for a metric (e.g. CPU = 50%) by adding/removing instances	Simplest; most common
Step Scaling	Triggers at alarm thresholds; scales by steps based on breach magnitude; does not wait for cooldown	Custom multi-step response
Simple Scaling	Single action per alarm; waits for cooldown before next action; legacy	Basic setups
Scheduled Scaling	Pre-define capacity changes for known patterns (e.g. scale up at 8 AM Mon–Fri)	Predictable load
Predictive Scaling	ML analyses historical patterns; provisions ahead of expected demand	Recurring traffic spikes

[!] Exam tip

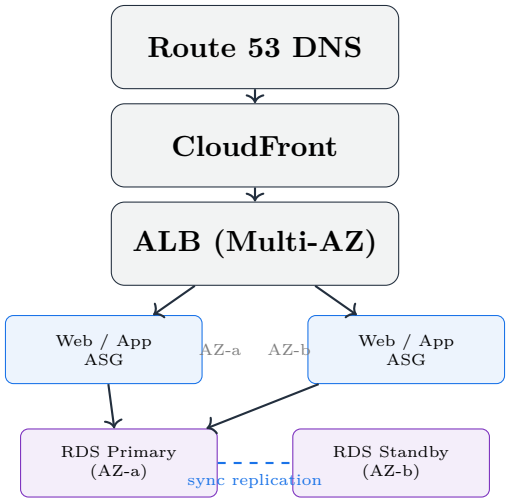
Target Tracking is the default recommendation for most scenarios. **Scheduled Scaling** is best when you know in advance (e.g. Black Friday launch, business-hours pattern). **Predictive** + Target Tracking together is the SAP-level answer for "automatically handle cyclical load without manual intervention".

7.2 Lifecycle Hooks

Pause instance launch or termination to perform custom actions:

- **Launch hook**: wait for instance to complete bootstrapping before serving traffic (e.g. pull config from S3, run software install). Timeout: 1 hour.
- **Termination hook**: drain connections, push logs to S3, deregister from discovery service before instance is terminated.

7.3 Multi-Tier Architecture Diagram



8 Elastic Load Balancing (ELB)

SAA+SAP *Distribute traffic across targets in one or more AZs*

[key] Key facts

- All ELB types are **highly available** by default (across AZs in the region).
- **Target types**: EC2 instances, IP addresses, Lambda functions, other ALBs (ALB only).
- **Health checks**: ELB removes unhealthy targets from rotation.
- **Cross-zone load balancing**: distributes traffic evenly across all registered targets regardless of AZ. Default on for ALB (cost per AZ for NLB/GWLB).
- **Sticky sessions (session affinity)**: route requests from same client to same target (ALB: cookie-based; NLB: source IP).

8.1 Load Balancer Types Comparison

ELB type selection

Type	Layer	Protocols	Best for
ALB	L7 (application)	HTTP, HTTPS, WebSocket	Host/path-based routing, microservices, containers, Lambda, gRPC
NLB	L4 (transport)	TCP, UDP, TLS	High-perf (millions req/s), gaming, IoT, static IP needed, PrivateLink
GWLB	L3 (gateway)	GENEVE (IP packets)	Inline inspection via 3rd-party virtual appliances (NGFW, IDS/IPS)
CLB	L4/L7	HTTP, HTTPS, TCP	Legacy; avoid for new workloads

[!] Exam tip

- **"Content-based routing"** (host header, URL path, HTTP method) ⇒ **ALB**.
- **"Static IP" or "millions of requests per second" or UDP ⇒ NLB**.
- **"Inspect all traffic through a firewall appliance" ⇒ GWLB**.
- NLB supports **PrivateLink** (expose service to other VPCs/accounts) — ALB does not.
- ALB supports **Lambda** as a target — NLB does not.

8.2 ALB Advanced Features

- **Listener rules**: route by host header (`api.example.com`), URL path (`/images/*`), query string, HTTP method, IP source. Priority-based evaluation.
- **Target groups**: pool of targets; each rule forwards to a target group. Can mix instance, IP, and Lambda targets.
- **Blue/green with ALB**: use weighted target groups to shift traffic gradually between old and new deployments.
- **Connection draining (deregistration delay)**: default 300 s; ALB allows in-flight requests to complete before removing a target.

[*] SAP depth

SAP: Auto Scaling and ELB at enterprise scale:

- **Mixed Instances Policy**: combine On-Demand and Spot Instances in a single ASG; specify multiple instance types. AWS rebalances Spot allocations to minimise interruptions.
- **Warm pools**: maintain pre-initialised instances in a "warm" stopped state;

reduce scale-out latency for apps with long boot times.

- **Scale-in protection**: mark specific instances as protected from scale-in (e.g. instances processing long jobs).
- **NLB + PrivateLink**: expose ALB or internal service to other AWS accounts/VPCs without VPC peering. NLB is the endpoint service; consumers create Interface Endpoints.
- **GWLB pattern**: VPC traffic → GWLB endpoint → GWLB → 3rd-party IDS/IPS (via GENEVE) → back to GWLB → destination. Deploy as a centralised inspection VPC (security VPC hub pattern).

9 Containers on AWS

SAP *ECS, EKS, Fargate, ECR — container orchestration*

[key] Key facts

- **Container**: packages application code + runtime + dependencies in a portable unit. Shares host OS kernel (lighter than VMs); starts in seconds.
- **Container image**: immutable snapshot; stored in a registry (ECR).
- **Orchestration**: manages scheduling, scaling, networking, and health of containers across a cluster.

9.1 Container Platform Selection

Container service comparison

Service	Description	Choose when...
Amazon ECS	AWS-native container orchestration; simpler than Kubernetes; tight AWS service integration	You want AWS-native simplicity; no Kubernetes expertise needed
Amazon EKS	Managed Kubernetes; AWS handles control plane	You need Kubernetes API compatibility, existing K8s tooling, or multi-cloud portability
AWS Fargate	Serverless compute engine for ECS or EKS; no EC2 instances to manage; pay per vCPU/memory/second	You want to run containers without managing servers
Amazon ECR	Managed container image registry; integrates with ECS, EKS, Lambda, CodeBuild; OCI-compliant	Always use ECR for images deployed to AWS services
ECS Anywhere	Run ECS tasks on on-premises servers using SSM agent	Hybrid; on-prem workloads with AWS control plane
EKS Anywhere	Run EKS on on-premises hardware (bare metal / VMware)	On-prem Kubernetes with EKS-compatible APIs
AWS App Runner	Fully managed; deploy container or source code; autoscales; no infrastructure knowledge needed	Simple web apps/APIs with no K8s/ECS expertise

[!] Exam tip

ECS vs EKS: ECS = simpler, AWS-native, less overhead. EKS = Kubernetes-compatible, more complex, use when K8s is a hard requirement (team expertise, multi-cloud, K8s ecosystem tools). **EC2 launch type vs Fargate:** EC2 = more control, GPU support, lower cost at sustained load. Fargate = serverless, no cluster management, pay-per-task (better for variable/bursty workloads).

9.2 Amazon ECS

- **Cluster**: logical grouping of tasks/services (EC2 or Fargate).
- **Task Definition**: blueprint specifying container image(s), CPU, memory, IAM role, networking mode, environment variables, log driver.
- **Task**: running instance of a task definition (one-time job).
- **Service**: maintains N running copies of a task; integrates with ALB/NLB; handles

rolling deployments and health checks.

- **Task IAM Role**: credentials for the container (fine-grained, per-task; prefer over EC2 instance role for containers).
- **Networking modes**: `awsvpc` (each task gets own ENI + security group — recommended), `bridge` (Docker NAT), `host`.

[*] SAP depth

ECS blue/green with CodeDeploy:

- CodeDeploy registers new task set; shifts ALB traffic from blue (old) to green (new) target group.
- Original task set retained for configurable rollback window (e.g. 1 hour).
- Automatic rollback triggers on CloudWatch alarm.
- Supports canary (10%→100%) or linear traffic shifting.

9.3 Amazon EKS

- Managed Kubernetes control plane (API server, etcd); AWS patches and scales it.
- **Node groups**: managed EC2 instances (Auto Scaling Group); or Fargate profiles for serverless nodes.
- **EKS Add-ons**: managed VPC CNI, CoreDNS, kube-proxy, EBS CSI driver.
- **IRSA (IAM Roles for Service Accounts)**: map K8s service accounts to IAM roles; fine-grained pod-level AWS permissions.
- **Karpenter**: open-source node provisioner for EKS; replaces Cluster Autoscaler; faster node provisioning, better bin-packing.

9.4 Amazon ECR

- Fully managed Docker-compatible container image registry.
- **Private repositories**: access via IAM (resource-based policies for cross-account).
- **Public Gallery**: host public images (`public.ecr.aws`).
- **Image scanning**: on-push or continuous scanning via Inspector; reports CVEs.
- **Lifecycle policies**: automatically delete untagged or old images to save storage cost.
- **Replication**: replicate images across regions and accounts for multi-region deployments.

9.5 Compute Platform Selection (Full Spectrum)

Compute platform decision guide

Platform	Choose when...
EC2	Full control needed; persistent workloads; custom OS; GPU; lift-and-shift
Elastic Beanstalk	PaaS; auto-manages infra (EC2, ALB, ASG); web apps and APIs; replatform
Lambda	Event-driven; short-duration; serverless; auto-scales to zero
ECS on Fargate	Containers; no cluster management; variable load
ECS on EC2	Containers; sustained high load; GPU workloads; cost optimisation
EKS	Kubernetes required; multi-cloud; K8s ecosystem tooling
App Runner	Simple web apps; source code or container; no infra knowledge
AWS Outposts	On-premises; data residency; hybrid; low-latency to on-prem systems

10 CI/CD Pipeline Services

SAP *CodeCommit, CodeBuild, CodeDeploy, CodePipeline*

[key] Key facts

- **CodeCommit**: managed Git repository; private, encrypted, integrated with IAM. (Note: AWS announced end-of-life for new customers in 2024 — GitHub/GitLab often preferred.)
- **CodeBuild**: fully managed build service; compiles source, runs tests, produces artifacts. Pay per build minute. No servers to manage.
- **CodeDeploy**: automates application deployments to EC2, on-premises servers, Lambda, and ECS. Supports rolling, blue/green, and canary deployment strategies.
- **CodePipeline**: orchestrates the full CI/CD workflow; connects source → build → test → deploy stages. Integrates with GitHub, CodeCommit, CodeBuild, CodeDeploy, CloudFormation, Elastic Beanstalk, and ECS.
- **CodeArtifact**: managed artifact repository for npm, Maven, PyPI, NuGet packages. Proxies public package registries with a security boundary.

[!] Exam tip

The standard SAP CI/CD pipeline is: **Source** (CodeCommit / GitHub) → **Build** (CodeBuild) → **Deploy** (CodeDeploy / CloudFormation / ECS) orchestrated by **CodePipeline**. Each stage can trigger an approval action (manual gate) before

proceeding.

10.1 Deployment Strategies

Deployment strategy comparison

Strategy	How it works	Downtime	Rollback
All-at-once (In-place)	Deploy to all instances simultaneously	Yes (brief)	Re-deploy old version
Rolling	Deploy to a subset at a time; old and new coexist	Minimal	Re-deploy old
Rolling + extra batch	Launch extra capacity first; then roll; maintains capacity	None	Re-deploy old
Immutable	Launch brand-new instances; replace old entirely	None	Terminate new
Blue/Green	New environment in parallel; switch traffic (DNS or ALB)	None	Flip DNS back
Canary	Shift small % of traffic to new version; monitor; increase	None	Route 100% back

[!] Exam tip

Blue/Green is the preferred SAP answer for "zero-downtime" + "easy rollback". Control traffic shifting with Route 53 weighted routing or ALB target group weights. CodeDeploy natively supports blue/green for EC2, Lambda (traffic shift), and ECS. **Canary** (e.g. 10% → 100%) is best for gradual validation with automated rollback on CloudWatch alarm.

[*] SAP depth

SAP: Advanced deployment patterns:

- **Lambda Canary/Linear**: CodeDeploy shifts Lambda traffic in configurable steps (e.g. `LambdaCanary10Percent5Minutes`: 10% for 5 min, then 100%). Automatic rollback on CloudWatch alarm.
- **ECS Blue/Green**: CodeDeploy registers a new task definition; shifts ALB traffic between blue (old) and green (new) target groups. Original task set is retained for rollback window.
- **CloudFormation Change Sets**: preview infra changes (like a diff) before applying. Use in CodePipeline for IaC deployments with human approval gates.
- **Service Catalog**: package CloudFormation templates as approved Products; developers self-service without direct CF access. Enforces governance at scale.

- **AWS Proton**: managed deployment service for container and serverless applications; platform teams define environment/service templates; dev teams self-deploy.

11 AWS Systems Manager (SSM)

SAA+SAP *Unified operational management for EC2 and hybrid infrastructure*

[key] Key facts

- **Agentless or agent-based**: SSM Agent pre-installed on Amazon Linux 2 / Windows AMIs. On-premises servers use the agent with a hybrid activation.
- No need to open SSH (port 22) or RDP (port 3389) for management operations.
- SSM communicates over HTTPS to the SSM endpoint (no inbound ports required).

11.1 Key SSM Capabilities

SSM features

Feature	Purpose
Session Manager	Interactive shell access to EC2 and on-premises without SSH/RDP. Session logs go to S3/CloudWatch. Replaces bastion hosts.
Parameter Store	Hierarchical key-value store for config and secrets. Standard tier free; Advanced tier supports larger values and policies (TTL).
Run Command	Execute commands at scale across a fleet of instances without SSH. Output goes to S3/CloudWatch.
Patch Manager	Define patch baselines (which patches are approved); schedule maintenance windows; scan and remediate.
State Manager	Define and maintain desired configuration state of instances (e.g. antivirus installed, port 22 closed).
Inventory	Collect metadata: installed software, OS config, network config. Query across fleet; feed into Config.
Automation	Runbooks (YAML/JSON) for automated remediation workflows (e.g. restart service, create snapshot, rotate key).
OpsCenter	Central location for operational issues (OpsItems); integrates with CloudWatch alarms, Config, GuardDuty findings.

Parameter Store vs Secrets Manager

Feature	Parameter Store	Secrets Manager
Cost	Free (Standard) / $0.05/10k API (Advanced)	$0.40/secret/month + API
Auto-rotation	No native rotation	Yes (built-in for RDS/Aurora)
Max size	4 KB (Standard) / 8 KB (Advanced)	64 KB
Cross-account	Via IAM / resource policy	Native cross-account
Versioning	Yes	Yes
Use for	Config values, non-sensitive params, simple secrets	DB passwords, API keys, auto-rotated credentials

[!] Exam tip

"Automatically rotate database credentials" ⇒ **Secrets Manager**. **"Store application configuration cheaply"** ⇒ **Parameter Store** (Standard tier is free). **"Replace bastion host"** ⇒ **SSM Session Manager** (no open ports, full audit trail).

Part IV

Storage (SAP depth)

12 Amazon S3

SAA+SAP *Object storage — 11 nines durability, unlimited capacity*

[key] Key facts

- Objects stored in **buckets**; bucket names are globally unique.
- Object size: 0 bytes to 5 TB; max single PUT is 5 GB. Use **Multipart Upload** for objects >100 MB.
- **Durability**: 99.999999999% (11 nines) — replicated across ≥3 AZs.
- **Availability**: varies by storage class (Standard = 99.99%).
- S3 is a **regional** service; bucket name namespace is global.
- **Strong consistency** (since Dec 2020): all operations are strongly consistent.
- Throughput: **3 500 PUT/COPY/DELETE** + **5 500 GET/HEAD** per prefix per second.

12.1 Storage Classes

S3 storage class quick reference

Class	Retrieval	Min days	Use case
Standard	Milliseconds	—	Frequently accessed data
Intelligent-Tiering	Milliseconds	—	Unknown / changing access patterns
Standard-IA	Milliseconds	30	Infrequent access, fast retrieval
One Zone-IA	Milliseconds	30	Reproducible infrequent data (single AZ)
Glacier Instant	Milliseconds	90	Archives needing instant access
Glacier Flexible	1 min – 12 h	90	Backup / archive, hours OK
Glacier Deep Archive	12 h	180	Long-term compliance, cheapest

[!] Exam tip

Retrieval fees apply to IA and Glacier classes. Glacier Deep Archive = cheapest storage but 12-hour retrieval — wrong for "retrievable within hours" scenarios (use Glacier Flexible). One Zone-IA loses data if that AZ fails — never for unique or non-reproducible data.

12.2 Security

- **Block Public Access**: account-level or bucket-level setting; overrides any bucket policy or ACL. Enable by default unless hosting public static sites.
- **Bucket Policies**: resource-based JSON IAM policies; can grant cross-account access.
- **ACLs**: legacy; AWS recommends disabling ACLs and using bucket policies instead.
- **Pre-signed URLs**: time-limited URL grants temporary access to a private object without changing bucket permissions. Generated by a principal with `s3:GetObject`.
- **VPC Gateway Endpoint**: route S3 traffic within VPC without NAT / internet gateway. Free; add to route tables. (Interface Endpoint = PrivateLink, costs per GB.)

12.2.1 Server-Side Encryption (SSE)

S3 encryption options

Type	Key managed by	Notes
SSE-S3	AWS (S3-managed AES-256)	Default; no extra cost
SSE-KMS	AWS KMS (CMK or AWS-managed)	Audit trail via CloudTrail; extra KMS API cost
SSE-C	Customer (you supply key per request)	AWS does not store key; HTTPS required
Client-side	Customer (encrypt before upload)	Full key control; more complex

[!] Exam tip

Default encryption: enable SSE-S3 or SSE-KMS on the bucket so all new objects are encrypted automatically. To *require* encryption, add a bucket policy that `Denys` PUT requests without the `x-amz-server-side-encryption` header.

12.3 Versioning & Replication

Versioning

- Once enabled, can only be *suspended*, never disabled.
- Stores all versions of an object (including deletes as delete markers).
- **MFA Delete**: requires MFA to permanently delete versions or change versioning state.

Replication

- **CRR** (Cross-Region Replication): DR, compliance, latency.
- **SRR** (Same-Region Replication): log aggregation, test/prod sync.
- Both require *versioning enabled* on source and destination buckets.
- Replication is *not retroactive*: existing objects are not replicated. Use **S3 Batch Operations** to replicate existing objects.
- Replication does not copy delete markers by default (configurable).

12.4 S3 Lifecycle Policies

Automate moving or deleting objects based on age:

- **Transition actions**: move to cheaper storage class after N days (e.g. Standard → Standard-IA after 30 days → Glacier after 90 days).
- **Expiration actions**: delete objects (or noncurrent versions) after N days.

- Minimum 30-day rule: objects must stay in Standard-IA / One Zone-IA ≥30 days before transitioning.

12.5 Performance & Transfer

- **Multipart Upload**: parallel upload of parts; required for >5 GB, recommended >100 MB. Increases throughput, recovers from network failures.
- **S3 Transfer Acceleration**: routes uploads/downloads through CloudFront edge locations over AWS backbone. Good for global uploads to a single bucket.
- **S3 Select / Glacier Select**: run SQL queries against CSV/JSON/Parquet objects in S3; retrieve subset of data (up to 400% faster, 80% cheaper than full retrieval).
- **Byte-range fetches**: download only a specific byte range; enables parallel multi-thread downloads and partial object retrieval.
- Prefix performance: spread objects across prefixes to maximise throughput (5 500 GET/prefix/s × many prefixes).

[i] SAA depth

SAA: S3 features for application architectures:

- **Static website hosting**: host HTML/CSS/JS directly from S3 (enable in bucket settings); pair with CloudFront for HTTPS and global CDN.
- **Event Notifications**: trigger SQS, SNS, Lambda, or EventBridge on `s3:ObjectCreated`, `s3:ObjectRemoved`, etc.
- **Object Lock (WORM)**: write-once-read-many; Governance mode (admins can override) vs Compliance mode (nobody can delete during retention, including root).
- **Requester Pays**: download costs billed to the requester, not bucket owner. Useful for data sets shared with external parties.
- **CORS**: configure cross-origin resource sharing so browser apps on domain A can fetch S3 objects hosted under domain B.

[*] SAP depth

SAP: S3 at enterprise scale:

- **S3 Access Points**: create named access points with their own policies for different applications or teams — simplifies bucket policy management at scale.
- **Multi-Region Access Points**: single global endpoint routes requests to the lowest-latency bucket; failover across regions; enables active-active S3.
- **Object Lambda**: attach a Lambda function to a `GetObject` call; transform data on the fly (redact PII, convert format) without storing multiple copies.
- **S3 as Data Lake**: backbone for Athena (serverless SQL), AWS Glue (ETL), Lake Formation (governance, fine-grained access). Store raw + processed tiers.
- **S3 Batch Operations**: run Lambda or S3-native operations (copy, tag, restore, invoke Lambda) across billions of objects; useful for retroactive replication or re-encryption after KMS key rotation.

- **Compliance Object Lock**: Compliance mode cannot be overridden by any user (including root); satisfies SEC 17a-4, FINRA, HIPAA retention requirements.

[!] Exam tip

Common S3 exam traps:

- CRR/SRR requires versioning on *both* buckets — not just the source.
- Replication does NOT copy existing objects — need S3 Batch Operations.
- S3 Gateway Endpoint is *free*; Interface Endpoint (PrivateLink) costs money.
- Glacier Deep Archive has 12-hour retrieval — not "a few hours" (use Flexible).
- **S3 is not a file system** — no append, no atomic file update, no lock. For shared file access use EFS; for block storage use EBS.

13 Shared File Storage & Storage Gateway

SAA+SAP *EFS, FSx, Storage Gateway — file systems and hybrid storage*

13.1 Storage Service Selection

AWS storage service comparison

Service	Type / Protocol	Use case / Choose when...
S3	Object store	Web data, backups, data lake, static files
EBS	Block (single-EC2)	OS disk, databases, high-IOPS single instance
EFS	NFS shared file (Linux)	Shared access from multiple EC2 / Lambda / ECS
FSx for Windows	SMB (Windows)	Windows workloads, Active Directory integration
FSx for Lustre	Parallel FS (POSIX)	HPC, ML training, burst compute from S3
FSx for NetApp	NFS, SMB, iSCSI	Lift-and-shift NetApp ONTAP workloads
FSx for OpenZFS	NFS (ZFS)	Linux apps using ZFS features; snapshot-heavy
Storage Gateway	File / Volume / Tape	Hybrid on-prem to AWS; extend on-prem storage

[!] Exam tip

"Multiple EC2 instances need shared access to the same files" ⇒ **EFS** (Linux) or **FSx for Windows** (Windows/SMB). **Never EBS** for shared access — EBS is attached to a single EC2 instance (except Multi-Attach io2, single AZ only).

13.2 Amazon EFS — Elastic File System

[key] Key facts

- Fully managed **NFS** file system; **Linux only** (NFSv4.1/4.2).
- **Elastic**: automatically grows and shrinks; no capacity planning.
- **Multi-AZ**: data stored across multiple AZs by default (Regional mode). One Zone mode (single AZ) is cheaper (∼47% less) for non-critical data.
- Mount from EC2, Lambda (via VPC), ECS, EKS.
- **Performance modes**: General Purpose (default, low latency); Max I/O (higher throughput, higher latency — large parallel workloads).
- **Throughput modes**: Bursting (scales with storage size); Provisioned (fixed throughput regardless of size); Elastic (auto-scales throughput).
- **Storage classes**: Standard (frequent access); Infrequent Access (IA, cheaper); Archive (rarely accessed, cheapest). Lifecycle Management moves files automatically.

13.3 Amazon FSx

13.3.1 FSx for Windows File Server

- Fully managed **SMB** (Windows file share) backed by Windows Server.
- Integrates with **Microsoft Active Directory** (on-premises AD or AWS Managed AD).
- Supports: DFS namespaces, DFS replication, User Quotas, ACLs.
- Multi-AZ deployment for high availability.
- On-premises access via Direct Connect or VPN.
- Equivalent on-prem: Windows File Server.

13.3.2 FSx for Lustre

- High-performance **parallel file system** for HPC, ML, media processing.
- **POSIX-compliant**; sub-millisecond latency; hundreds of GB/s throughput; millions of IOPS.
- Seamless integration with **S3**: lazy-load data from S3 on first access; write results back to S3. Ideal for burst-out ML training from a data lake.
- **Deployment types**: Scratch (short-term, no replication, cheapest); Persistent (long-term, replicated within AZ).
- On-premises access via Direct Connect or VPN.

[!] Exam tip

"HPC / ML training / high-throughput parallel access from many nodes" ⇒ FSx for Lustre. Millions of IOPS at sub-millisecond latency; pairs with S3 as the data source.

13.3.3 FSx for NetApp ONTAP

- Managed NetApp ONTAP; supports **NFS, SMB, iSCSI**.
- Best for lifting and shifting workloads that use NetApp on-premises.
- Features: deduplication, compression, thin provisioning, snapshots, SnapMirror replication.
- Multi-AZ with automated failover.

13.3.4 FSx for OpenZFS

- Managed ZFS file system; **NFS** access.
- For Linux applications that rely on ZFS features: snapshots, clones, compression.
- Up to 1 million IOPS; sub-millisecond latency.

13.4 EBS Volume Types

EBS volume type selection

Type	Category	Max IOPS	Use case
gp3	General SSD	16 000	Most workloads; configurable IOPS/throughput (decoupled from size)
gp2	General SSD	16 000	Legacy general purpose (prefer gp3 for new volumes)
io2	Provisioned SSD	256 000	High-performance DBs; Multi-Attach; SAP HANA
io1	Provisioned SSD	64 000	Legacy high-IOPS (prefer io2)
st1	Throughput HDD	N/A	Big data, log processing, sequential large reads (max 500 MB/s)
sc1	Cold HDD	N/A	Infrequently accessed; lowest cost block storage

[!] Exam tip

"Highest IOPS possible" ⇒ io2 Block Express (256K IOPS). **"Cost-effective general SSD" ⇒ gp3** (cheaper than gp2, independent IOPS/throughput tuning). **Cannot encrypt an existing unencrypted volume directly** — create a snapshot, copy with encryption enabled, then create new volume from encrypted snapshot.

13.5 AWS Storage Gateway

[key] Key facts

- Hybrid storage service bridging on-premises applications to AWS storage.
- Runs as a VM appliance on-premises (VMware, Hyper-V, EC2) or hardware appliance.

Storage Gateway modes

Mode	How it works	Use case
File Gateway	NFS/SMB interface; files stored as objects in S3; local cache for frequently accessed data	On-prem apps that need file access but store data in S3
Volume Gateway	iSCSI block volumes backed by S3 + EBS snapshots. *Cached mode*: primary data in S3, cache on-prem. *Stored mode*: primary on-prem, async backup to S3	iSCSI block storage with cloud backup / DR
Tape Gateway	Virtual Tape Library (VTL); iSCSI interface; virtual tapes in S3 / Glacier	Replace physical tape backups; use existing backup software

[*] SAP depth

SAP: Storage migration strategies:

- **AWS DataSync**: automate and accelerate data transfer from on-prem NAS/NFS/SMB to S3, EFS, or FSx. Handles scheduling, encryption, integrity checks. Up to 10 Gb/s per task.
- **AWS Transfer Family**: managed SFTP, FTP, FTPS, AS2 server fronting S3 or EFS. Replace on-prem FTP servers without changing existing client workflows.
- **Storage Gateway File mode**: use as a long-term bridge; on-prem apps see NFS while data lives in S3. Good for gradual migration to cloud.
- **Snow Family → S3**: when network transfer would take >1 week, use Snowball

Edge (terabytes) or Snowmobile (petabytes) for offline bulk transfer.

Part V

Networking (SAP depth)

14 Amazon VPC

SAA+SAP *Your isolated virtual network within AWS — the foundation of all networking.*

14.1 Core Components

[key] Key facts

- **CIDR block** — IPv4 /16 to /28; IPv6 /56 (optional)
- **Subnets** — within one AZ; public (route to IGW) vs private
- **Internet Gateway (IGW)** — allows public subnet ↔ internet
- **Route Tables** — one per subnet (or shared); most specific route wins
- **NAT Gateway** — lets private subnets reach internet (outbound only). Managed, **Cost note:** charged per GB + per hour. AZ-specific; deploy one per AZ for HA.
- **NACL** — subnet-level, stateless firewall; rules evaluated in order
- **Security Group** — instance-level, stateful; no explicit deny (only allow)

14.2 VPC Connectivity

14.2.1 VPC Endpoints

14.2.2 VPC Peering

14.2.3 Transit Gateway (TGW)

14.3 DNS in VPC

[i] SAA depth

[*] SAP depth

[!] Exam tip

[!!] Gotcha

VPC Peering is **not transitive**: if VPC A peers with B, and B peers with C, A cannot reach C through B. Use **Transit Gateway** for hub-and-spoke connectivity across many VPCs.

15 Hybrid & VPC Connectivity

SAA+SAP *Connecting VPCs, on-premises networks, and services securely.*

15.1 VPC Connectivity Options

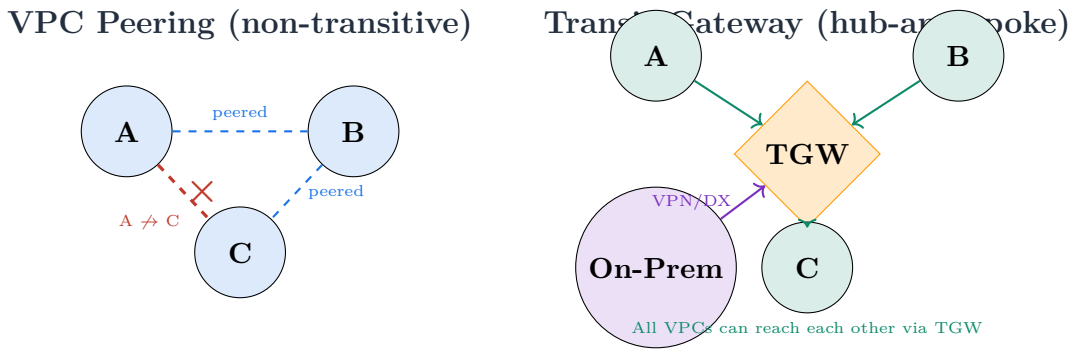


Figure 1: Left: VPC Peering is not transitive (A cannot reach C via B). Right: Transit Gateway enables any-to-any routing including on-premises.

VPC Connectivity Options

Method	Description	Key Constraint
VPC Peering	Direct link between two VPCs; non-transitive	No overlapping CIDRs; scales poorly at >20 VPCs
Transit Gateway (TGW)	Hub-and-spoke; transitive; cross-account; cross-region	Per-attachment + per-GB cost
AWS PrivateLink	Expose a service to other VPCs without peering; traffic on AWS backbone	One-directional; consumer → provider
VPC Endpoints (Gateway)	Free private route to S3 and DynamoDB via route table	S3 and DynamoDB only
VPC Endpoints (Interface)	PrivateLink ENI in subnet; for most other AWS services	Per-AZ cost

[!!] Gotcha

VPC Peering is NOT transitive. If A peers with B and B peers with C, A cannot reach C via B. This is the most frequently tested VPC concept. Use **Transit Gateway** when you need any-to-any connectivity across many VPCs or on-premises

networks.

15.2 Hybrid Connectivity: Direct Connect & VPN

Hybrid connectivity options

Service	Description	Speed / Note
AWS Direct Connect	Dedicated private connection from on-premises to AWS; consistent low latency	1, 10, 100 Gbps
Site-to-Site VPN	Encrypted IPsec tunnel over public internet; fast to provision	Up to 1.25 Gbps
Client VPN	Managed OpenVPN for remote user access	Per-user
Direct Connect + VPN	VPN over Direct Connect for encryption; combines privacy and encryption	Production pattern

[!] Exam tip

Resilient hybrid connectivity pattern: use **Direct Connect as primary** (consistent, low-latency) and **VPN as failover**. Direct Connect alone is not redundant — provision two DX connections from different providers or pair with VPN. For maximum resilience: two DX connections in different locations.

[i] SAA depth

For SAA:

- Direct Connect takes **weeks** to provision; VPN is **minutes**.
- Use VPC **Gateway Endpoints** for S3/DynamoDB access (free, no data processing charge vs NAT Gateway).
- Use VPC **Interface Endpoints** (PrivateLink) for other AWS services to avoid traffic leaving the VPC.

[*] SAP depth

SAP networking depth:

- **Transit Gateway + Direct Connect Gateway** — connect a single TGW to multiple regions and on-premises over DX; use transit VIF (Virtual Interface).
- **Shared VPC via AWS RAM** — share subnets from a central network account to spoke accounts; centralised routing without peering mesh.
- **Centralised egress via TGW + NAT Gateway** — single NAT GW in a dedicated egress VPC; all spoke VPCs route through TGW to the NAT GW.
- **AWS Network Firewall** — deploy in centralised inspection VPC; route all ingress/egress through it via TGW and Gateway Load Balancer.

- **Direct Connect SLA:** 99.9% (1 connection) or 99.99% (2 connections from different locations). Design for the SLA you need.

15.3 Hybrid DNS with Route 53 Resolver

[key] Key facts

- **Inbound Resolver Endpoint** — allows on-premises DNS servers to resolve AWS private hosted zone records (on-prem → AWS DNS)
- **Outbound Resolver Endpoint** — allows VPC instances to forward DNS queries to on-premises DNS (AWS → on-prem DNS)
- **Forwarding Rules** — define which domains are forwarded to on-premises resolvers
- Deploy at least two endpoints across AZs for high availability

[!] Exam tip

On the SAP exam: hybrid DNS questions typically involve a company that has on-premises DNS (e.g., Active Directory) that needs to resolve AWS private hostnames (VPC DNS) and vice versa. The answer is always **Route 53 Inbound + Outbound Resolver Endpoints** with forwarding rules — not a custom DNS on EC2.

16 Amazon Route 53

SAA+SAP *Managed DNS, domain registration, health checks, traffic routing*

[key] Key facts

- **Authoritative DNS** service; manages DNS records for your domains.
- Supports domain registration (purchase/transfer domains through Route 53).
- **Hosted Zones**: container for DNS records. Public (internet-facing) or Private (resolves within a VPC).
- **Health Checks**: monitor endpoints, other health checks (calculated), or CloudWatch alarms. Used to enable DNS failover.
- **Alias records**: Route 53-specific extension; points to AWS resources (ALB, CloudFront, Elastic Beanstalk, S3 website, another Route 53 record). *No TTL cost*; resolves at Route 53 edge; can target zone apex (`example.com`).
- **CNAME vs Alias**: CNAME cannot be used at zone apex; Alias can.

16.1 Routing Policies

Route 53 routing policies

Policy	How it works	Typical use case
Simple	Returns all values; client picks randomly	Single resource; no health checks
Weighted	Distributes traffic by percentage weight	A/B testing, blue/green deployments
Latency	Routes to region with lowest network latency for the user	Multi-region active-active
Failover	Primary → Secondary on health-check failure	Disaster recovery, active-passive
Geolocation	Routes by user's geographic location (continent, country, state)	Content localisation, data residency
Geoproximity	Routes by geographic distance, adjustable with bias	Shift traffic between regions (Traffic Flow)
Multi-Value	Returns up to 8 healthy records; client picks	Simple load distribution with health checks
IP-based	Routes by CIDR block of the client's IP	ISP optimisation, cost reduction

[!] Exam tip

- **Failover** = active-passive DR with automatic DNS cutover.
- **Latency** = "lowest latency for the user" = multi-region active-active.
- **Geolocation** = route by country/continent — data sovereignty.
- **Weighted** = A/B test or gradual traffic shift (e.g. 10%/90% blue-green).
- **Multi-Value** is NOT a load balancer replacement (no sticky sessions, no health-aware connection pooling) — use ALB for that.

16.2 Health Checks

- Monitor endpoints over HTTP, HTTPS, or TCP.
- **Calculated health checks**: combine multiple sub-checks (AND/OR logic).
- **CloudWatch alarm health check**: triggers based on a CloudWatch metric (e.g. SQS queue depth, custom metric) rather than an HTTP endpoint.
- Health checks originate from Route 53 health checkers in multiple regions (must allow traffic from Route 53 CIDR ranges).

16.3 Private Hosted Zones

- Resolves internal DNS names within one or more VPCs.

- VPC must have **DNS Resolution** and **DNS Hostnames** enabled.
- Associate multiple VPCs to one private hosted zone (cross-account via CLI/API).
- For hybrid (on-prem to AWS DNS): use **Route 53 Resolver inbound / outbound endpoints** (see hybrid_connectivity.tex).

[*] SAP depth

SAP: Route 53 advanced patterns:

- **Traffic Flow + Geoproximity**: visual editor for complex routing; shift traffic between regions with bias value (−99 to +99).
- **Resolver DNS Firewall**: block queries to malicious domains; complements Route 53 Resolver for VPC-level DNS filtering.
- **DNSSEC**: sign hosted zone records to prevent DNS spoofing; supported for public hosted zones.
- **Multi-region active-active**: Latency routing + health checks + Global Accelerator (for non-HTTP workloads). Route 53 handles DNS; Global Accelerator handles anycast IP.
- **Failover chain**: Route 53 Failover → CloudFront → ALB → EC2/ECS in primary region; on primary failure, DNS flips to secondary region ALB.

17 Amazon CloudFront

SAA+SAP *Global CDN — cache content at edge, reduce origin load*

[key] Key facts

- **Content Delivery Network (CDN)**: serves cached content from 400+ edge locations worldwide; reduces latency for static and dynamic content.
- **Origins**: S3 bucket, ALB, EC2, API Gateway, custom HTTP server.
- **Distributions**: Web (HTTP/HTTPS) for websites and APIs.
- **TTL**: configurable per behaviour; cache-control headers honoured.
- **Protocol**: HTTP/HTTPS; enforces TLS; integrates with AWS Certificate Manager.
- **Geo Restriction**: whitelist or blacklist countries.
- **Signed URLs / Signed Cookies**: restrict access to authenticated users (e.g. paid media, software downloads).
- **Price Classes**: select which edge locations to use to reduce cost.

17.1 Origin Access Control (OAC)

To serve S3 content exclusively through CloudFront (block direct S3 access):

1. Create an **OAC** (Origin Access Control) — the modern replacement for OAI.
2. Attach OAC to the CloudFront distribution for the S3 origin.

3. Update the S3 bucket policy to allow `s3:GetObject` from the OAC principal.
4. Keep **Block Public Access** enabled on the S3 bucket.

[!] Exam tip

OAC is the current best practice for S3 + CloudFront. OAI (Origin Access Identity) is the legacy option — some exam questions still reference it. Both work similarly but OAC supports SSE-KMS encrypted buckets.

17.2 Lambda@Edge & CloudFront Functions

Edge compute options

Feature	Lambda@Edge	CloudFront Functions
Runtime	Node.js, Python	JavaScript
Triggers	Viewer / origin request & response	Viewer request & response only
Max exec time	5 s (viewer) / 30 s (origin)	1 ms
Memory	128 MB – 10 GB	2 MB
Use cases	A/B testing, JWT auth, URL rewrite, personalisation	Simple URL rewrites, header manipulation, cache key normalisation

17.3 CloudFront vs AWS Global Accelerator

CloudFront vs Global Accelerator

Factor	CloudFront	Global Accelerator
Layer	L7 (HTTP/HTTPS CDN)	L4 (TCP/UDP) or L7
Caching	Yes — primary value	No caching
Protocols	HTTP/HTTPS, HLS	TCP, UDP, HTTP
Static IPs	No (DNS-based)	Yes — 2 static anycast IPs
Use cases	Static assets, web APIs, streaming	Gaming, IoT, VoIP, non-HTTP, multi-region ALB
DDoS protection	AWS Shield Standard	AWS Shield Standard

[!] Exam tip

- **Static files / website acceleration** ⇒ CloudFront (caching).
- **"Static IP for global traffic"** ⇒ Global Accelerator (2 static anycast IPs).
- **UDP or non-HTTP protocols** ⇒ Global Accelerator.

- **"Speed up API responses without caching"**: both can work; Global Accelerator routes over AWS backbone vs CloudFront intelligent routing — they are similar for dynamic, uncacheable content.

[i] SAA depth

SAA: Common CloudFront architecture patterns:

- **S3 static site + CloudFront**: S3 hosts HTML/CSS/JS → CloudFront for HTTPS, global caching, and OAC to keep S3 private.
- **Dynamic + static**: single CloudFront distribution with multiple behaviours: `/api/*` → ALB (no cache); `/static/*` → S3 (long TTL).
- **Signed URLs**: distribute paid media or software. URL per-file; short expiry. Signed Cookies: multiple files per session (e.g. streaming subscription).

[*] SAP depth

SAP: CloudFront advanced:

- **Origin Groups / Failover**: primary origin (ALB region A) fails → CloudFront automatically retries secondary origin (ALB region B). DR with DNS-level failover.
- **Field-Level Encryption**: encrypt specific POST fields (e.g. credit card) at the edge before sending to origin; only designated backend can decrypt.
- **Cache Policies and Origin Request Policies**: fine-grained control of what goes into the cache key (headers, query strings, cookies) and what is forwarded to origin.
- **Real-Time Logs**: stream access logs to Kinesis Data Streams for real-time analytics.
- **Continuous Deployment**: test new CloudFront configurations on a traffic sample before promoting to production; zero-downtime config rollout.

Part VI

Security & Identity (SAP depth)

18 AWS IAM

SAA+SAP *Control who can do what on which AWS resources*

[key] Key facts

- **Users**: long-term credentials (console password + access keys). One per human.
- **Groups**: collection of users; attach policies to groups, not directly to users.
- **Roles**: assumed by users, services, or federated identities; short-lived temporary

credentials via STS. No static keys.

- **Policies**: JSON documents defining Allow/Deny on Actions/Resources/Conditions.
- **Global service**: IAM is not region-specific; applies across all regions.
- **Principle of Least Privilege**: grant only what is explicitly needed.

18.1 IAM 10 Security Best Practices

1. **Lock away root user access keys** — never create keys for root; use only for account-level tasks (e.g. change root email). Enable MFA on root.
2. **Create individual IAM users** — one identity per human; avoid shared accounts.
3. **Use groups to assign permissions** — assign policies to groups; add users to groups.
4. **Grant least privilege** — start with minimal permissions; expand only as needed. Use IAM Access Analyzer to identify overly broad policies.
5. **Enable MFA for all users** — especially for privileged/admin users.
6. **Use IAM roles for applications on EC2** — never store access keys on instances; use instance profiles for temporary credentials.
7. **Use roles to delegate access** — cross-account access via IAM roles, not sharing credentials.
8. **Rotate credentials regularly** — rotate access keys periodically; use Secrets Manager for automatic rotation of service credentials.
9. **Remove unnecessary credentials** — disable/delete inactive users and unused keys. Use credential report (`aws iam generate-credential-report`) to audit.
10. **Monitor activity with CloudTrail** — enable a multi-region trail; alert on sensitive API calls (e.g. `DeleteBucket`, `CreateUser`).

18.2 Policy Types

IAM policy types

Policy type	Description and attachment point
Identity-based	Attached to IAM users, groups, or roles. Inline (embedded) or managed (standalone).
Resource-based	Attached to a resource (S3 bucket, SQS queue, KMS key, Lambda). Specifies who (principal) can access. Enables cross-account.
Permission boundary	IAM-managed policy set as the maximum permissions boundary for a user or role; cannot grant *more* than the boundary allows.
SCP	AWS Organizations policy; limits what accounts in an OU can do. Does not grant permissions — sets the ceiling.
Session policy	Inline policy passed during `AssumeRole`; further limits the session's permissions. Used in federation.

18.2.1 Policy Evaluation Logic

Evaluation order (all must allow; any explicit deny wins):

1. **SCP** — if deny, stop. If no allow statement, implicitly deny.
2. **Permission Boundary** — limits effective permissions of user/role.
3. **Identity-based policy** — must explicitly allow the action.
4. **Resource-based policy** — if resource policy allows same account principal, implicit allow (resource and identity policies work together for same-account).
5. **Explicit Deny anywhere** — overrides all allows (highest priority).

[!] Exam tip

Default is Deny. All actions are implicitly denied unless explicitly allowed. An explicit `Deny` in any policy always wins, even if another policy has an `Allow`. **SCPs do not grant permissions** — an SCP allow without a corresponding IAM allow does nothing.

18.3 IAM Roles & STS

- **AssumeRole**: a principal calls STS `AssumeRole`; receives temporary security credentials (access key, secret key, session token) valid 15 min – 12 hours.
- **Cross-account roles**: account A trusts account B in the role's trust policy; users in B can assume the role in A. Cleanest cross-account access pattern.
- **Instance Profiles**: IAM role attached to an EC2 instance; SDK on the instance automatically retrieves temporary credentials from IMDS (`169.254.169.254`).

- **External ID**: condition in the trust policy for third-party cross-account access; prevents the *confused deputy problem* (third party uses your role for another customer).
- **Role chaining**: assuming a role that then assumes another role; max session duration resets to 1 hour; can cause complexity — use sparingly.

18.4 IAM Identity Center (SSO)

- Centralised access management for multiple AWS accounts and cloud applications.
- Replaces legacy AWS SSO; integrates with Active Directory or external IdPs (Okta, Azure AD).
- **Permission Sets**: templates applied to accounts; map to IAM roles in each account.
- Single sign-on portal: users log in once, access all assigned accounts and apps.
- Recommended replacement for individual IAM users for human access.

18.5 MFA & Credential Security

- **MFA types**: Virtual (Google Authenticator), Hardware TOTP token, FIDO2/WebAuthn (hardware security key), SMS (not recommended).
- **Credential report**: CSV listing all IAM users, their credential types, last-used, rotation dates, MFA status. Generate with `aws iam generate-credential-report`.
- **Access Advisor**: shows which services a user or role has accessed and when. Helps identify and remove unused permissions.

[i] SAA depth

SAA: IAM for EC2 and S3 patterns:

- EC2 → S3: use instance profile (IAM role), not access keys. Role policy grants `s3:GetObject`; no credentials stored on instance.
- S3 cross-account: bucket policy (resource-based) grants the other account's role permission; *and* the role's identity policy must allow `s3:*`. Both must allow — resource policy alone is not enough for same-account but is sufficient for cross-account when the bucket policy grants access.
- **IAM Access Analyzer**: continuously analyses resource policies to find resources shared outside the account or organisation.

[*] SAP depth

SAP: IAM at enterprise scale:

- **SCPs + IAM together**: SCPs are guardrails on the OU; IAM policies grant individual access. Use SCPs to deny risky actions (disable CloudTrail, leave Org, create IAM users) across all member accounts without touching individual account IAM.
- **Permission Boundaries**: delegate IAM administration safely. Developer can create roles but only up to the boundary — prevents privilege escalation.

- **ABAC (Attribute-Based Access Control)**: use resource/request tags as IAM conditions. One policy covers many resources if tags match. Scales better than RBAC for large teams.
- **SAML 2.0 federation**: enterprise IdP (AD, Okta) → STS `AssumeRoleWithSAML` → temporary AWS credentials. Users authenticate to corporate IdP, not directly to AWS.
- **OIDC federation**: web/mobile apps use Google/Facebook/Cognito IdP → STS `AssumeRoleWithWebIdentity`.

[!] Exam tip

SCP vs IAM policy: SCPs set the *maximum* permissions for an account in an AWS Organization — they do *not* grant permissions themselves. An SCP `Allow` alone does nothing; the IAM policy must also allow the action.

[!!] Gotcha

Confused deputy attack: when a third-party assumes a role in your account, always use an **External ID** condition in the trust policy to prevent unauthorised cross-account access.

19 AWS KMS — Key Management Service

SAA+SAP *Managed encryption key creation, control, and auditing*

[key] Key facts

- **Customer Master Keys (CMKs)** now called **KMS Keys**:
  - *AWS-managed keys*: created/managed by AWS for specific services (e.g. `aws/s3`). Free. Cannot rotate manually. Shared key per service per account.
  - *Customer-managed keys (CMK)*: you create and control. $1/month + API call costs. Configurable rotation (annual automatic or manual on-demand).
  - *AWS-owned keys*: owned by AWS services; no visibility or control.
- **Envelope encryption**: CMK encrypts a data encryption key (DEK); DEK encrypts actual data. Only DEK is stored encrypted alongside the data.
- **Key Policies**: resource-based policy on the KMS key; must explicitly allow IAM principals (even account root). Without a key policy, no one can use the key.
- **CloudTrail integration**: every KMS API call is logged (who used which key, when).
- **Multi-Region Keys**: replicate keys across regions for cross-region decryption (Aurora Global, DynamoDB Global Tables, S3 CRR with SSE-KMS).

[!] Exam tip

KMS key types exam decision:

- Need **audit trail** of key usage ⇒ Customer-managed CMK (CloudTrail logs all).
- Need to **bring your own key material** ⇒ CMK with imported key material.
- Need **FIPS 140-2 Level 3** (not just Level 2) ⇒ **CloudHSM**.
- **Rotate automatically** ⇒ enable automatic annual rotation on CMK (AWS keeps old versions for decryption of existing ciphertext).

[*] SAP depth

SAP: KMS advanced:

- **Cross-account**: grant another account permission in the key policy; the other account's IAM policy must also allow it (two-step authorisation).
- **KMS Grants**: temporary delegated permissions for specific operations (e.g. EC2 Auto Scaling uses grants to encrypt EBS volumes).
- **CloudHSM**: dedicated HSM cluster (FIPS 140-2 Level 3); you manage keys entirely. No AWS access. Integrates with KMS as custom key store.

20 AWS Shield & WAF

SAA+SAP *DDoS protection and web application firewall*

20.1 AWS Shield

Shield Standard vs Advanced

Feature	Shield Standard	Shield Advanced
Cost	Free (automatic)	$3 000/month per organisation
Protection	L3/L4 DDoS (volumetric)	L3/L4 + L7 (sophisticated)
Coverage	All AWS customers	EC2, ELB, CloudFront, Route 53, Global Accelerator
DRT access	No	Yes — AWS DDoS Response Team (24/7)
Cost protection	No	Yes — credits for spikes due to DDoS
Advanced detection	No	Yes — real-time attack visibility, reports

20.2 AWS WAF — Web Application Firewall

[key] Key facts

- Protects at **Layer 7 (HTTP/HTTPS)**; attaches to ALB, API Gateway, CloudFront, or AppSync.
- **Web ACLs**: contain rules that `Allow`, `Block`, or `Count` requests.
- **Rule types**: IP set, geographic, rate-based, string/regex match, SQL injection, XSS, size constraints.
- **AWS Managed Rules**: pre-built rule groups (OWASP Top 10, AWSManagedRulesCommonRuleSet, AWSManagedRulesKnownBadInputsRuleSet).
- **Rate-based rules**: block IPs exceeding N requests per 5 minutes — effective for brute force and credential stuffing.

[!] Exam tip

WAF is **Layer 7** only (HTTP/HTTPS). For Layer 3/4 DDoS ⇒ Shield. For WAF: must attach to ALB, CloudFront, or API Gateway — cannot attach to NLB directly. Rate-based rules are the go-to for "block if more than N requests per IP".

21 Additional Security Services

SAA+SAP *GuardDuty, Inspector, Macie, Security Hub, Secrets Manager*

Security service decision table

Service	What it does / When to use
GuardDuty	ML-based threat detection; analyses VPC Flow Logs, CloudTrail, DNS logs. Finds: compromised EC2, crypto mining, unusual API calls. Agentless; enable with one click.
Inspector	Automated vulnerability scanning for EC2 (OS/package CVEs), Lambda functions, and container images in ECR. Continuous; agent on EC2.
Macie	ML-based PII/sensitive data discovery in S3. Identifies SSN, credit cards, API keys in object content.
Security Hub	Aggregates findings from GuardDuty, Inspector, Macie, IAM Access Analyzer, Firewall Manager. Single security dashboard. CSPM.
Secrets Manager	Stores, rotates, and retrieves secrets (DB passwords, API keys). **Auto-rotates** credentials (built-in for RDS/Aurora). Replaces hard-coded credentials.
Systems Manager Parameter Store	Cheaper secret/config store; hierarchical key-value. No auto-rotation. Use for non-sensitive config + simple secrets.

[!] Exam tip

- **"Automatically rotate database password"** ⇒ **Secrets Manager**.
- **"Detect if EC2 is mining cryptocurrency"** ⇒ **GuardDuty**.
- **"Find PII in S3 buckets"** ⇒ **Macie**.
- **"Scan EC2 for CVEs"** ⇒ **Inspector**.
- **"Centralise security findings across accounts"** ⇒ **Security Hub**.

[*] SAP depth

SAP: Security at enterprise scale:

- **AWS Firewall Manager**: centrally manage WAF rules, Shield Advanced, Security Groups, and Network Firewall policies across all accounts in an AWS Organization.
- **IAM Access Analyzer**: identifies resources shared externally (S3 buckets, IAM roles, KMS keys, SQS queues). Generates least-privilege policies from CloudTrail activity.
- **Detective**: investigate security incidents; visualises relationships between GuardDuty findings, CloudTrail logs, VPC Flow Logs.
- **Certificate Manager (ACM)**: provision and auto-renew TLS certificates for ALB, CloudFront, API Gateway, Elastic Beanstalk. Free for public certificates.

Part VII

Databases (SAP depth)

22 Amazon RDS & Aurora

SAA+SAP *Managed relational databases. RDS for standard engines; Aurora for AWS-native performance.*

22.1 Amazon RDS

[key] Key facts

- Supported engines: MySQL, PostgreSQL, MariaDB, Oracle, SQL Server, Db2
- AWS manages: patching, backups, hardware, failover
- **Multi-AZ** — synchronous standby in another AZ; automatic failover (1–2 min); standby is *not* readable (except Multi-AZ cluster mode)
- **Read Replicas** — asynchronous; up to 5 per source; can be cross-region. Promote to standalone if needed.
- **Automated backups** — 1–35 day retention; point-in-time restore
- Storage: gp2/gp3/io1; auto-scaling storage available

22.1.1 Multi-AZ vs Read Replicas

22.2 Amazon Aurora

22.2.1 Aurora Global Database

22.2.2 Aurora Serverless v2

22.3 RDS Proxy

22.4 Security

[i] SAA depth

[*] SAP depth

[!] Exam tip

Read Replica for DR: you can promote a cross-region Read Replica to a standalone DB, making it a valid (though async) DR option. For near-zero RPO/RTO, use **Aurora Global Database**.

[!!] Gotcha

Encrypting an existing unencrypted RDS instance is **not** possible directly. You must: take a snapshot → copy snapshot with encryption → restore from encrypted snapshot.

23 Amazon DynamoDB

SAA+SAP *Fully managed, serverless NoSQL key-value and document store*

[key] Key facts

- **Key structure**: Partition Key (required) + optional Sort Key. Primary key = Partition Key alone, or Partition Key + Sort Key (composite).
- **Capacity modes**:
  - *On-Demand*: no capacity planning; pay per request. Best for unpredictable traffic.
  - *Provisioned*: specify Read Capacity Units (RCU) and Write Capacity Units (WCU). Can add Auto Scaling; cheaper for predictable workloads.
- **Performance**: single-digit millisecond latency at any scale.
- **Item size**: max 400 KB per item.
- **Durability**: 99.999999999% (11 nines); data replicated across 3 AZs.
- **DynamoDB Accelerator (DAX)**: in-memory cache; microsecond read latency; drop-in compatible with DynamoDB API.

23.1 DynamoDB vs RDS — When to Choose

DynamoDB vs RDS comparison

Factor	DynamoDB	RDS (Relational)
Data model	Key-value / document (schema-less)	Tables with fixed schema, foreign keys
Query style	Get/put by key; limited scan; GSI for alternate access patterns	Full SQL: joins, aggregations, complex queries
Scale	Horizontal; unlimited items	Vertical (instance size) + read replicas
Latency	Single-digit ms at any scale	Low ms; varies with query complexity
Transactions	DynamoDB Transactions (ACID, same region)	Full ACID with SQL
Operational	Fully managed (serverless option)	Managed but you choose instance type
Best for	User sessions, shopping carts, gaming leaderboards, IoT events	E-commerce orders, financial records, reports needing joins

[!] Exam tip

If the question says "**flexible schema**", "**millisecond latency at scale**", "**serverless**", or "key-value" — DynamoDB. If the question says "**complex joins**", "**relational data**", "existing SQL application", or "ACID transactions with SQL" — RDS / Aurora.

23.2 Secondary Indexes

- **Global Secondary Index (GSI)**: different partition key (and optional sort key) than the table. Spans all partitions. Can be created/deleted at any time. Eventual consistency.
- **Local Secondary Index (LSI)**: same partition key, different sort key. Must be created at table creation. Supports strongly consistent reads. Limited to 10 GB per partition.

23.3 DynamoDB Streams & Triggers

- **DynamoDB Streams**: ordered stream of item-level changes (insert/modify/delete). Retention: 24 hours. Consumed by Lambda (event source mapping) for real-time processing.
- **Use cases**: audit logs, replication to other stores, triggering downstream workflows.

23.4 DynamoDB Transactions

- **TransactWriteItems / TransactGetItems**: atomic, consistent, isolated, durable. Up to 100 items or 4 MB per transaction.

- Useful for: banking (debit/credit atomically), inventory management, shopping cart checkout.

23.5 Time To Live (TTL)

Define an attribute as TTL; DynamoDB auto-deletes expired items (within 48 hours, eventually). **Free**. Use for: session data, temporary tokens, logs, IoT data.

[i] SAA depth

SAA: DynamoDB in application patterns:

- **Session store**: stateless web servers store user sessions in DynamoDB; TTL removes stale sessions automatically.
- **Caching pattern**: use DAX in front of DynamoDB for read-heavy workloads (product catalogue, leaderboards). DAX provides microsecond reads.
- **Event-driven**: DynamoDB Streams → Lambda for change-data-capture (CDC) — fan-out to search indexes, caches, analytics.
- **Serverless stack**: API Gateway → Lambda → DynamoDB; no servers to manage, scales to zero, pay-per-request.

[*] SAP depth

SAP: DynamoDB enterprise patterns:

- **Global Tables**: multi-region, active-active replication; sub-second latency worldwide; automatic conflict resolution (last-writer-wins). Requires on-demand capacity or provisioned with auto-scaling.
- **Point-in-Time Recovery (PITR)**: continuous backup; restore to any second in the last 35 days. Enable for production tables.
- **Export to S3**: export full table snapshots to S3 for analysis in Athena / Glue without consuming read capacity.
- **Capacity planning**: 1 RCU = 1 strongly consistent read / 2 eventually consistent reads of items ≤4 KB. 1 WCU = 1 write of items ≤1 KB. Hot partitions ("partition key anti-patterns") limit throughput — spread writes across many partition keys (e.g. add random suffix / shard key).

24 Purpose-Built Databases

SAA+SAP *Right database for the right workload*

AWS offers specialised databases for different data models. Choosing the right one is a core exam skill.

Purpose-built database selection guide

Service	Model / Use case	Key characteristic
RDS / Aurora	Relational (SQL); OLTP; structured data with joins	ACID; up to 5× MySQL, 3× PostgreSQL performance (Aurora)
DynamoDB	Key-value / document; any scale; millisecond latency	Serverless; TTL; Global Tables; DAX caching
Redshift	OLAP / data warehouse; petabyte-scale analytics	Columnar; Redshift Spectrum queries S3 directly
ElastiCache	In-memory caching (Redis or Memcached)	Sub-millisecond latency; sessions, leaderboards
Neptune	Graph database; highly connected data	Social networks, fraud detection, knowledge graphs
DocumentDB	MongoDB-compatible; JSON document model	Managed MongoDB; auto-scales storage
Keyspaces	Apache Cassandra-compatible; wide-column	Serverless; scales automatically; Cassandra CQL
Timestream	Time-series; IoT, operational metrics, telemetry	Stores trillions of events/day; built-in analytics
OpenSearch	Search & log analytics; full-text search	Elasticsearch-compatible; Kibana dashboards
MemoryDB	Redis-compatible; durable in-memory primary store	Persistence with multi-AZ; microsecond reads

[!] Exam tip

Quick decision triggers:

- "Graph / social network / fraud detection" ⇒ **Neptune**
- "Time-series / IoT telemetry" ⇒ **Timestream**
- "Full-text search / log analytics" ⇒ **OpenSearch**
- "MongoDB workload" ⇒ **DocumentDB**
- "Cassandra workload" ⇒ **Keyspaces**
- "Petabyte analytics / data warehouse" ⇒ **Redshift**
- "Durable Redis (primary database, not just cache)" ⇒ **MemoryDB**
- "Session cache / leaderboard / pub-sub" ⇒ **ElastiCache (Redis)**

24.1 Amazon Redshift

[key] Key facts

- Petabyte-scale **columnar data warehouse**; OLAP (not OLTP).
- **Redshift Spectrum**: run SQL queries directly on data in S3 without loading; extends warehouse to data lake.
- **Concurrency Scaling**: automatically adds temporary read capacity during bursts.
- **RA3 nodes**: separate compute and storage; store data on S3-backed managed storage (lower cost at scale).
- Integrates with QuickSight, Athena, SageMaker, Glue.
- Single AZ by default; **Multi-AZ** option available for RA3 (preview).

[!!] Gotcha

Redshift is for **analytics (OLAP)**, not transactions (OLTP). Never use Redshift for high-concurrency transactional workloads — use Aurora or RDS instead. INSERT/UPDATE performance is poor compared to row-store databases.

24.2 Amazon ElastiCache

ElastiCache Redis vs Memcached

Feature	Redis	Memcached
Data structures	Strings, hashes, lists, sets, sorted sets, streams	Simple key-value only
Persistence	RDB snapshots + AOF log	None
Replication	Primary + replicas; Multi-AZ failover	None
Cluster mode	Yes (horizontal sharding)	Yes (multi-node)
Pub/Sub	Yes	No
Sorted sets	Yes (leaderboards, rate limiting)	No
Lua scripting	Yes	No
Best for	Sessions, leaderboards, pub/sub, real-time analytics, complex data structures	Simple distributed caching, multi-threaded, stateless

Caching patterns:

- **Cache-aside (lazy loading)**: app checks cache first; on miss, fetches from DB and populates cache. Most common pattern. Handles cache misses gracefully.

- **Write-through**: write to cache and DB simultaneously. Cache always current; extra write penalty; may cache data never read.
- **Session store**: store user sessions in Redis with TTL; stateless app servers; ALB sticky sessions not needed.

[!] Exam tip

Redis for anything requiring persistence, replication, complex data structures, pub/sub. **Memcached** for simple, multi-threaded, stateless caching where simplicity matters. In practice, Redis is almost always the right answer on the exam.

24.3 Amazon OpenSearch Service

[key] Key facts

- Managed Elasticsearch + Kibana (OpenSearch Dashboards); fork from Elasticsearch 7.10.
- **Use cases**: application log analytics, full-text search, real-time monitoring, clickstream analytics.
- Common pattern: data pipeline → Kinesis Firehose → OpenSearch; or CloudWatch Logs → Lambda → OpenSearch.
- Multi-AZ with 3+ data nodes; dedicated master nodes for stability.
- **UltraWarm**: lower-cost storage tier for infrequently queried data.
- **Cold Storage**: detach indices to S3; reattach for querying.

[*] SAP depth

SAP: Database modernisation patterns:

- **Oracle/SQL Server → Aurora**: eliminate expensive licensing costs. Use SCT + DMS for schema conversion and migration. Aurora Babelfish allows SQL Server apps to run on Aurora PostgreSQL with minimal code change.
- **MySQL/PostgreSQL → Aurora**: drop-in compatible; up to 5×/3× performance.
- **Self-managed Elasticsearch → OpenSearch Service**: eliminate operational burden.
- **Relational → DynamoDB**: for simple key-value access patterns at scale; eliminates provisioned capacity management.
- **Data Warehouse → Redshift**: for petabyte-scale OLAP; pair with S3 data lake (Redshift Spectrum) and Glue ETL.
- **DAX in front of DynamoDB**: drop-in cache; microsecond reads; no application code changes beyond pointing to DAX endpoint instead of DynamoDB.

Part VIII

Integration, Messaging & Analytics

25 Messaging & Application Integration

SAA+SAP *Decouple producers and consumers; event-driven architectures; workflow orchestration.*

25.1 Messaging Decision Framework

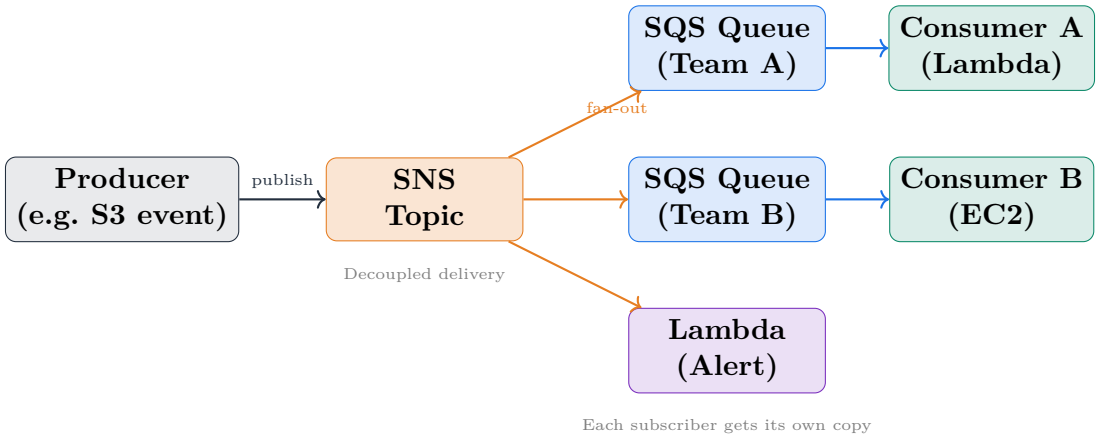


Figure 2: SNS → SQS fanout pattern: one message published to SNS fans out to multiple SQS queues, each processed independently at their own pace.

25.2 Amazon SQS

[key] Key facts

- **Standard Queue** — at-least-once delivery; best-effort ordering; nearly unlimited throughput
- **FIFO Queue** — exactly-once processing; strict order; up to 3,000 msg/s with batching
- **Visibility Timeout** — message hidden after receive; default 30 s; max 12 hours; consumer deletes on success
- **Dead-Letter Queue (DLQ)** — messages that fail processing after `maxReceiveCount` are moved here for analysis
- **Long Polling** — reduce empty receives; wait up to 20 s for messages (preferred over short polling)
- Message size: max **256 KB**; retention: up to **14 days**

[!] Exam tip

SQS as a buffer: when a downstream service (e.g. EC2 processing fleet) cannot keep up with bursts, add an SQS queue between the producer and consumer. The queue absorbs the burst; consumers drain at their own rate. This is the canonical

"decouple with least operational overhead" pattern.

25.3 Amazon SNS

[key] Key facts

- **Pub/sub** — publish once, deliver to many subscribers simultaneously
- Subscribers: SQS queues, Lambda, HTTP/HTTPS endpoints, email, SMS, mobile push
- **Message Filtering** — per-subscription filter policies reduce unnecessary processing
- **FIFO Topics** — ordered, deduplicated delivery to SQS FIFO queues only
- **Fan-out pattern**: SNS → multiple SQS queues for parallel, independent processing

25.4 Amazon EventBridge

[key] Key facts

- **Serverless event bus** — route events from AWS services, custom apps, and 200+ SaaS partners
- **Event rules** — match events by pattern and route to targets (Lambda, SQS, Step Functions, API GW, etc.)
- **Scheduled rules** — cron or rate expressions for periodic triggers (replaces CloudWatch Events)
- **Event Archive & Replay** — store events and replay them for debugging or re-processing
- **Cross-account event routing** — publish events from one account's bus to another's

SQS vs SNS vs EventBridge — when to use each

Service	Best for	Consumer model
SQS Standard	Work queues; rate limiting; decoupling tiers	Pull; one consumer per message
SQS FIFO	Ordered, deduplicated processing (financial transactions)	Pull; one consumer per message
SNS	Fan-out to multiple subscribers simultaneously	Push; all subscribers get copy
EventBridge	Rule-based routing; SaaS integration; event replay	Push; rule-matched targets
Amazon MQ	Lift-and-shift of ActiveMQ/RabbitMQ workloads	Existing MQ protocol support
Kinesis	Real-time streaming data; ordered; replay; analytics	Pull; multiple consumers (shards)

[!] Exam tip

Classic combo: SNS → SQS = fan-out to queues for parallel, independently-scaled processing. Each team/microservice gets its own SQS queue and processes at its own pace without blocking others.

25.5 AWS Step Functions

[key] Key facts

- Orchestrate **microservices** and Lambda functions into serverless workflows (state machines)
- **Standard workflows** — long-running (up to 1 year); exactly-once; audit history; charged per state transition
- **Express workflows** — short-duration (<5 min); at-least-once; high volume; cheaper
- Built-in retry, catch, and error handling — eliminates custom retry logic in code
- Integrates with Lambda, ECS, DynamoDB, SQS, SNS, API GW, Bedrock, and more

[i] SAA depth

For SAA: understand SQS/SNS fan-out and EventBridge scheduled rules for serverless architectures. Know DLQ for failed message handling.

[*] SAP depth

At SAP level:

- **Strangler Fig with EventBridge:** gradually replace a monolith by routing

traffic through EventBridge rules — new microservices handle specific event types while the monolith handles the rest.

- **Amazon MQ** — use when migrating existing ActiveMQ or RabbitMQ; not for new architectures (use SQS/SNS instead for lower ops overhead).
- **Step Functions for orchestration:** prefer Step Functions over Lambda-orchestrates-Lambda patterns (avoids hidden retry complexity, cost, and timeouts).
- **EventBridge Pipes:** connect event sources to targets with optional filtering, enrichment, and transformation (reduces glue Lambda code).
- Cross-account: share an EventBridge event bus via resource-based policy; enables org-wide event routing without custom infrastructure.

26 Analytics & Streaming

SAA+SAP *Service selection, data lake patterns, and streaming pipelines.*

26.1 Analytics & Streaming Service Reference

Analytics services — purpose and memory hook

Service	Purpose	Memory Hook
Amazon Kinesis Data Streams	High-throughput real-time ingestion and processing; ordered per shard; replay up to 7 days; multiple consumers	Streaming ingest
Amazon Data Firehose	Fully managed delivery to S3, Redshift, or OpenSearch; 60 s buffer or 1 MB; no consumer code needed	Stream to storage
Amazon MSK (Managed Kafka)	Managed Apache Kafka; use when you need Kafka-native tooling or Kafka Connect	Managed Kafka
Amazon EMR	Managed Hadoop/Spark cluster; big data processing, ETL, ML at scale; pay for EC2 or Serverless	Big data processing
Amazon Redshift	Petabyte-scale columnar data warehouse; OLAP analytics; Redshift Spectrum queries S3 in place	Analytics warehouse
Amazon Athena	Interactive SQL directly on S3; serverless; pay per query; no ETL or loading required	SQL on S3
AWS Glue	Serverless ETL service; prepares/transforms data; Data Catalog stores metadata; used in data lake pipelines	ETL / data integration
Amazon OpenSearch Service	Full-text search, log analytics, near-real-time dashboards; successor to Elasticsearch	Search & log analytics
Amazon QuickSight	Managed BI dashboards and visual analytics; connects to Athena, Redshift, S3, RDS	BI dashboards

26.2 Kinesis Deep Dive

Kinesis Data Streams vs Data Firehose

	Kinesis Data Streams	Data Firehose
Management	You manage shards/scaling	Fully managed; auto-scales
Consumers	Multiple independent consumers	One destination at a time
Latency	~70–200 ms	Near-real-time (60 s or 1 MB buffer)
Replay	Up to 7 days	No replay
Processing	Custom Lambda / KCL consumer	Built-in transform (Lambda); no code needed
Ordering	Ordered per shard	Not guaranteed
Use when	Multiple consumers, replay, ordering needed	Deliver to S3/Redshift/OpenSearch fast

[key] Key facts

- **Shard capacity**: 1 MB/s write, 2 MB/s read per shard; scale out by splitting shards.
- **Enhanced fan-out**: dedicated 2 MB/s per consumer per shard; use when multiple consumers need high throughput independently.
- **SQS vs Kinesis**: SQS = decouple, buffer, fire-and-forget; Kinesis = ordered, replay, analytics on a continuous stream.
- **MSK vs Kinesis**: MSK when you need Kafka-native APIs, Kafka Connect, or Kafka Streams; Kinesis when you want fully managed AWS-native integration.

26.3 Analytics Service Decision Tree

[key] Key facts

- **Ad-hoc SQL on S3, no infrastructure → Athena** (pay per query; works on Parquet/ORC/JSON/CSV directly in S3).
- **Ongoing SQL analytics at petabyte scale → Redshift** (load data; columnar; fast joins; Spectrum for S3 overflow).
- **Serverless ETL, schema detection, data catalog → AWS Glue** (Glue Crawlers auto-detect schema; Glue Jobs transform with PySpark/Python).
- **Full-text search, log ingestion, dashboards → OpenSearch** (ingest from Firehose or Logstash; Kibana/OpenSearch Dashboards).
- **BI dashboards, business analytics → QuickSight** (SPICE in-memory engine; ML Insights; connects to Redshift/Athena/S3).
- **Big data Spark/Hadoop processing → EMR** (use Spot for task nodes to cut costs; persistent vs transient cluster).

- **Stream to S3/Redshift automatically** → **Firehose** (optionally transform with Lambda; buffer by time or size).
- **Real-time stream, multiple consumers, replay** → **Kinesis Data Streams**.

[!] Analytics exam traps

- **"No ETL, query data in place"** → Athena. *Not* Redshift (Redshift requires loading data into the cluster unless using Spectrum).
- **"Real-time log monitoring and alerting"** → Kinesis → Lambda *or* Firehose → OpenSearch. CloudWatch Logs Insights works too for AWS-native logs.
- **"Existing Kafka"** → MSK (migrate existing) or Kinesis (new build).
- **"Data lake with access control at column/row level"** → AWS Lake Formation on top of S3 + Glue Data Catalog.

[*] SAP depth

SAP analytics depth: SAP-C02 adds Lake Formation governance, EMR on EKS, Redshift RA3 nodes (separate compute/storage), Glue DataBrew (no-code transformation), Kinesis Analytics (SQL/Flink on streams), and multi-account data lake patterns using AWS RAM to share Glue Data Catalog resources across accounts. Cross-account Athena federation (Athena Federated Query) allows querying sources beyond S3 — RDS, DynamoDB, Redshift, CloudWatch — in a single SQL query.

Part IX

Reliability & DR (SAP depth)

27 Disaster Recovery & High Availability

SAA+SAP *RTO/RPO trade-offs and DR strategy selection across the cost/complexity spectrum.*

27.1 Key Reliability Metrics

[key] Key facts

- **RTO (Recovery Time Objective)** — maximum acceptable downtime after a failure. Lower RTO = faster recovery = higher cost.
- **RPO (Recovery Point Objective)** — maximum acceptable data loss, measured in time. Lower RPO = more frequent replication = higher cost.
- These metrics drive architecture decisions: always ask the business for RTO/RPO *before* choosing a DR strategy.

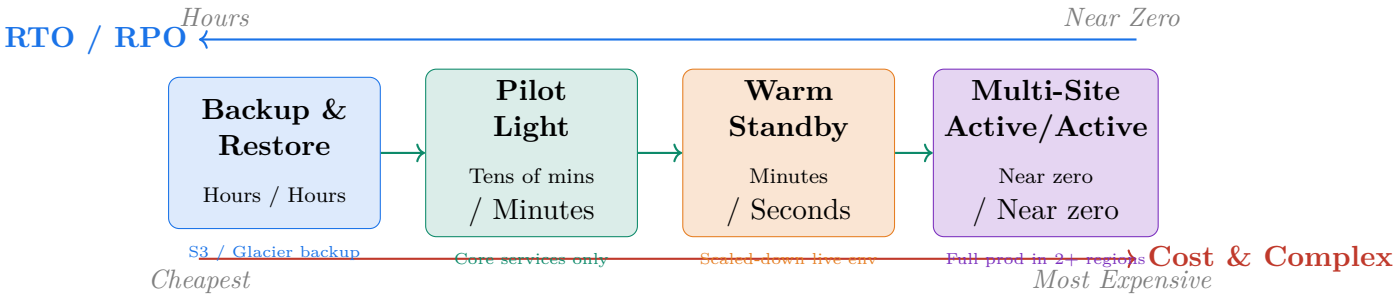


Figure 3: DR Strategy Spectrum — RTO/RPO in each box; cost increases left to right.

27.2 Disaster Recovery Strategies

DR Strategies

Strategy	Description	RTO	RPO
Backup & Restore	Data backed up to S3/Glacier; restore from scratch	Hours	Hours
Pilot Light	Minimal core (DB replicated); scale up on failover	Tens of mins	Minutes
Warm Standby	Scaled-down but fully functional environment	Minutes	Seconds
Multi-Site Active/Active	Full production in multiple regions simultaneously	Near zero	Near zero

[!] Exam tip

AWS Elastic Disaster Recovery (DRS) automates continuous block-level replication from on-prem or AWS to a target region. Use it for low-RTO/RPO without running full Warm Standby infrastructure. Backup & Restore is cheapest. Multi-Site is most expensive. *The exam gives you RTO/RPO requirements — map them to a strategy.*

[i] SAA depth

For SAA, focus on:

- RDS Multi-AZ for intra-region HA (synchronous standby, auto failover, same region)
- Cross-Region Read Replicas — promote to standalone for DR in another region (async)
- S3 Cross-Region Replication (CRR) for object-level DR
- Route 53 health checks for DNS-based failover

[*] SAP depth

At SAP level, expect to choose the *right* strategy for given RTO/RPO/cost constraints:

- **Aurora Global Database** — RPO < 1 s; RTO < 1 min. Primary region serves writes; up to 5 secondary regions for reads and DR promotion.
- **DynamoDB Global Tables** — multi-region, multi-active; built-in replication; sub-second RPO.
- **AWS Backup** — centralised backup across services (EC2, EBS, RDS, DynamoDB, EFS, FSx) with cross-account and cross-region copy.
- Pilot Light pattern: keep a minimal "always-on" config (IAM roles, DNS records, DB replica running); on disaster, scale up compute rapidly using pre-baked AMIs or CloudFormation.
- **AWS Resilience Hub** — continuously validates RTO/RPO against application architecture; recommends improvements.

27.3 High Availability Patterns

[key] Key facts

- **Multi-AZ deployments** — protects against AZ failure (RDS Multi-AZ, ALB across AZs, Auto Scaling across AZs)
- **Auto Scaling Groups** — automatically replace unhealthy instances; scale with demand; always span 2+ AZs
- **Elastic Load Balancing** — distributes traffic; health checks; cross-zone load balancing (ALB default ON; NLB default OFF)
- **Route 53 health checks** — DNS failover to healthy endpoints; can trigger CloudWatch alarm on failure
- **SQS decoupling** — decouple components so failure of one does not cascade (producer keeps writing; consumer retries)

[!!] Gotcha

ALB vs NLB cross-zone: ALB distributes evenly across all targets in all enabled AZs by default (cross-zone ON). NLB and GLB have cross-zone load balancing *off* by default — enable it explicitly if you have uneven instance counts per AZ.

Part X

Migration & Modernisation

28 Workload Migration & Modernisation

SAA+SAP *Strategy selection (7Rs), tools, and database migration patterns.*

28.1 The 7Rs Migration Strategies

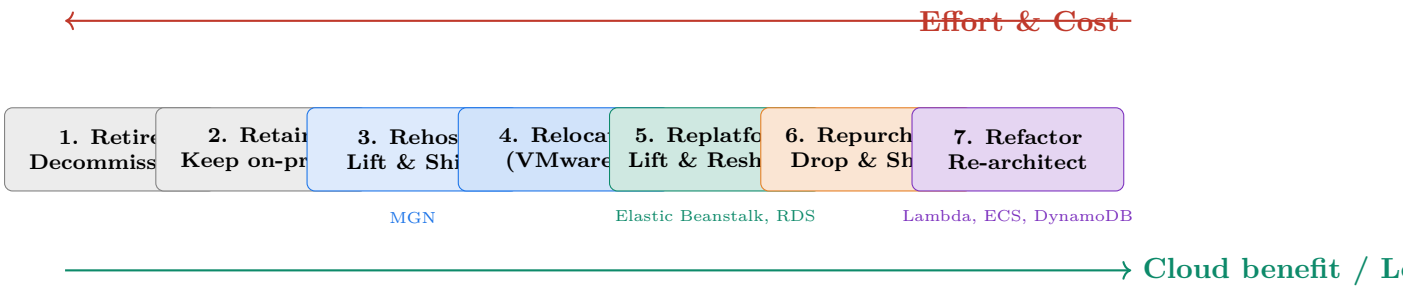


Figure 4: The 7Rs migration strategies ordered by increasing effort and cloud benefit.

7Rs Quick Reference

Strategy	Also Known As	When to Use
Retire	Remove	App no longer needed; decommission
Retain	Keep	Too risky, not worth migrating; revisit later
Rehost	Lift & Shift	Fast migration; no code changes; use MGN
Relocate	Lift & Shift	VMware on-prem to VMware Cloud on AWS
Replatform	Lift & Reshape	Minor optimisations (MySQL on EC2 → RDS); minimal code changes
Repurchase	Drop & Shop	Replace with SaaS (on-prem CRM → Salesforce)
Refactor	Re-architect	Max cloud benefit; rethink as microservices/serverless; highest cost

[!] Exam tip

Rehost = fastest, lowest effort. Refactor = most expensive, best long-term benefit. Do not overlook **Retire** and **Retain** — they are valid answers when the question mentions "budget constraints" or "app will be sunsetted within 6 months."

28.2 Application Migration Tools

Migration tools

Tool	Purpose
AWS Application Migration Service (MGN)	Lift-and-shift; continuous block-level replication from any source; minimal-downtime cutover
AWS Migration Hub	Central tracking of migration progress across all tools
AWS Application Discovery Service	Collect on-premises server inventory; agentless or agent-based
AWS DMS (Database Migration Service)	Migrate databases; homo- and heterogeneous; CDC for minimal downtime
AWS SCT (Schema Conversion Tool)	Convert source schema to target for heterogeneous migrations
AWS DataSync	High-speed incremental data transfer over network (NFS, SMB, S3, EFS)
AWS Transfer Family	SFTP, FTP, FTPS, AS2 server transferring to S3 or EFS
AWS Snow Family	Physical devices for offline bulk data transport
S3 Transfer Acceleration	Speed up S3 uploads using CloudFront edge locations

28.3 Snow Family

Snow Family devices

Device	Description	Capacity
Snowcone	Smallest; portable; edge computing + data transfer; rugged	8–14 TB
Snowball Edge	Storage-Optimised (80 TB) or Compute-Optimised (ML at edge)	80 TB usable
Snowmobile	Semi-truck for exabyte-scale migrations; AWS manages logistics	Up to 100 PB

[!] Exam tip

Rule of thumb: if transferring the data over the network would take *more than 1 week*, use Snow Family. For petabytes, use Snowmobile. For terabytes with edge compute needs, use Snowball Edge. For ongoing incremental sync (not bulk), use DataSync.

28.4 Database Migration Patterns

[key] Key facts

- **Homogeneous migration** (e.g. MySQL → RDS MySQL, Oracle → Oracle) — use **DMS** directly; no schema conversion needed
- **Heterogeneous migration** (e.g. Oracle → Aurora PostgreSQL) — use **SCT** first to convert schema, then **DMS** for data
- **CDC (Change Data Capture)** — DMS supports ongoing replication after initial load; minimises cutover downtime
- **Large database offline load** — use DMS + Snowball Edge for initial bulk load, then switch to CDC over the network for the delta

[i] SAA depth

SAA: know Snow Family selection (Snowcone vs Snowball vs Snowmobile by size), DMS for database migrations, DataSync for file/NFS/SMB transfers.

[*] SAP depth

SAP: know the full 7Rs and when each applies. Understand wave planning (group workloads by dependencies into migration waves). Key modernisation patterns:

- Oracle/SQL Server → Aurora (reduce licensing cost; Babelfish for SQL Server compat with Aurora PostgreSQL)
- MySQL/PostgreSQL → Aurora (up to 5× MySQL, 3× PostgreSQL performance)
- Self-managed Elasticsearch → Amazon OpenSearch Service
- Data warehouse → Amazon Redshift (petabyte-scale analytics, columnar storage)
- Strangler Fig pattern: gradually replace monolith functionality with microservices routing traffic via API Gateway or EventBridge

Part XI

Governance & Cost Management

29 AWS Organizations & Multi-Account Strategy

SAP *Central management of multiple AWS accounts; governance, logging, and resource sharing at scale.*

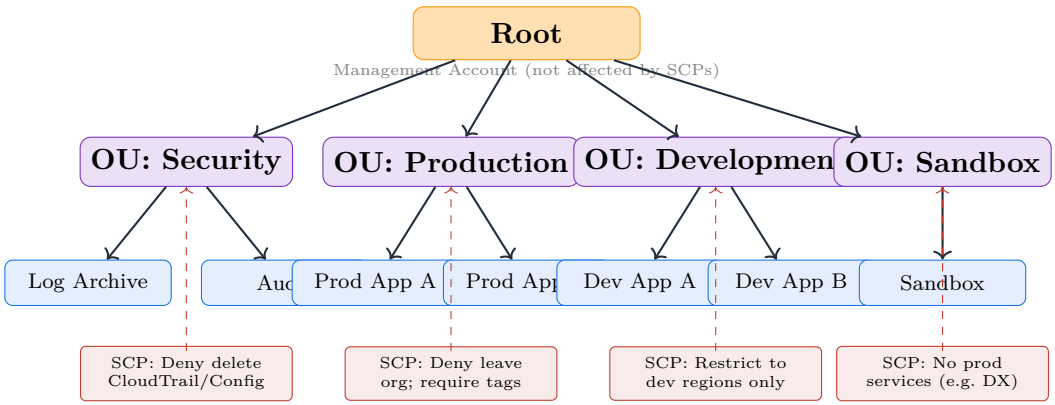


Figure 5: Sample AWS Organizations OU structure. SCPs flow down from Root through OUs to accounts.

29.1 AWS Organizations

[key] Key facts

- **Management Account** (formerly Master) — root of the org; creates/invites accounts; **NOT** affected by SCPs
- **Organizational Units (OUs)** — logical groupings; can be nested; SCPs apply to all accounts in OU
- **Service Control Policies (SCPs)** — set *maximum* permission boundaries; do NOT grant permissions; must be combined with IAM policies
- **Consolidated Billing** — aggregate usage for volume discounts; single payment method; RI/Savings Plan sharing across accounts

[!!] Gotcha

SCPs do NOT grant permissions — they only restrict the maximum permissions available. If an SCP has an explicit Deny, not even the account root user can perform that action. If an SCP has Allow, the IAM policy in the account must also Allow the action before it succeeds.

29.1.1 SCP Example: Restrict to Specific Regions

```
{
  "Effect": "Deny",
  "NotAction": ["iam:*", "support:*", "sts:*"],
  "Resource": "*",
  "Condition": {
    "StringNotEquals": {
      "aws:RequestedRegion": ["us-east-1", "ap-southeast-1"]
    }
  }
}
```

NotAction is used so IAM/STS (global services) still work regardless of region.

29.2 AWS Control Tower

[key] Key facts

- **Landing Zone** — a well-architected, multi-account baseline environment set up automatically
- **Guardrails** — pre-configured governance rules:
  - *Preventive* guardrails = SCPs (e.g. disallow root user access keys)
  - *Detective* guardrails = AWS Config rules (e.g. detect public S3 buckets)
- **Account Factory** — automates new account provisioning with approved baseline configurations; built on Service Catalog
- **Log Archive Account** — centralised CloudTrail and Config logs for all accounts
- **Audit Account** — read/write access to all accounts for security and compliance teams

[!] Exam tip

Organizations vs Control Tower: AWS Organizations is the underlying service; Control Tower is an opinionated, automated layer on top. If you need to set up a multi-account environment quickly with best practices baked in, Control Tower is the answer. If you need fine-grained custom control, use Organizations + SCPs directly.

29.3 Multi-Account Logging Strategy

[key] Key facts

- Centralise **CloudTrail** logs to a dedicated *Log Archive* account S3 bucket; enable org-level trail from management account
- Use **S3 bucket policies** to allow writes (from member accounts) but *deny deletes* — prevents tampering
- Enable **AWS Config** with an aggregator to view compliance across all accounts from a single pane
- Use **Amazon EventBridge** for cross-account event routing and notifications
- **AWS Security Hub** — aggregate findings from GuardDuty, Inspector, Macie, Config across all accounts; delegate admin to security account

29.4 AWS Resource Access Manager (RAM)

[key] Key facts

- Share AWS resources across accounts *without VPC peering*
- Shareable resources include: VPC subnets, Transit Gateway, Route 53 Resolver rules, License Manager configurations, Capacity Reservations
- **Shared VPC pattern:** central networking account owns the VPC and subnets; spoke accounts deploy resources into those subnets via RAM

- Does not copy resources — the resource stays in the owner account; RAM grants permissions

29.5 Cost Management Tools

Cost management tool comparison

Tool	Purpose
Cost Explorer	Visualise, understand, forecast AWS costs and usage; RI/SP recommendations
AWS Budgets	Set custom cost/usage/RI budgets with alert actions (apply SCP, stop EC2)
Cost and Usage Report	Most granular billing data; integrates with Athena + QuickSight
Trusted Advisor	Recommendations for cost, security, performance, fault tolerance
Compute Optimizer	ML-based rightsizing for EC2, ECS, Lambda, EBS
S3 Storage Lens	Org-wide S3 visibility; activity and usage analytics

29.5.1 Tagging Strategy for Cost Allocation

- Tag all resources: `CostCenter`, `Project`, `Environment` (prod/dev/staging), `Owner`
- **Activate cost allocation tags** in Billing Console to see them in Cost Explorer
- Use **AWS Config rules** or **SCPs** to enforce mandatory tagging on resource creation
- **AWS Budgets with actions** — automatically apply an SCP or stop EC2 instances when budget thresholds are breached

30 Amazon CloudWatch

SAA+SAP *Monitoring, metrics, logs, alarms, and dashboards*

[key] Key facts

- **Metrics**: time-series data points from AWS services and custom sources. Default resolution: 1-minute (detailed monitoring) or 5-minute (basic). **High-resolution**: up to 1-second resolution for custom metrics.
- **Alarms**: evaluate a metric over a period; trigger actions (SNS, Auto Scaling, EC2 action). States: OK, ALARM, INSUFFICIENT_DATA.
- **Logs**: collect, store, and search log files from EC2 (via CloudWatch Agent), Lambda, API Gateway, ECS, CloudTrail, VPC Flow Logs, etc.
- **Log Groups / Log Streams**: logs organised into groups (one per application/service) and streams (one per instance/Lambda invocation).

- **Log Insights**: ad-hoc query language for CloudWatch Logs. Analyse log data, calculate statistics, identify errors.
- **Dashboards**: cross-region, cross-account visualisations. Share with stakeholders.
- **Events / EventBridge**: schedule or respond to AWS events (CloudWatch Events is now EventBridge; same underlying service).

30.1 CloudWatch Agent

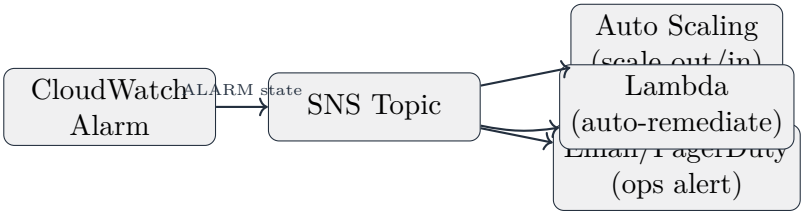
- Installed on EC2 (or on-premises) to ship custom metrics and logs to CloudWatch.
- **Default metrics** (no agent): CPU, network, disk I/O.
- **Requires agent**: Memory utilisation, disk space, process metrics.
- Config stored in Systems Manager Parameter Store for centralised management.

[!] Exam tip

"Monitor **memory** on EC2" ⇒ must install CloudWatch Agent. Default EC2 metrics do *not* include memory or disk space.

30.2 CloudWatch Alarms — Monitoring Chain

A common pattern: metric → alarm → SNS notification → Auto Scaling / Lambda / ops team.



[i] SAA depth

SAA: CloudWatch in application architectures:

- **Composite Alarms**: combine multiple alarms with AND/OR; reduces alarm noise. (E.g. alert only if both CPU AND memory are high.)
- **Log metric filters**: extract metric values from log text (e.g. count "ERROR" occurrences); create alarm on that custom metric.
- **Container Insights**: detailed metrics for ECS, EKS, K8s (requires agent).
- **Lambda Insights**: detailed function performance metrics (init duration, etc.).

[*] SAP depth

SAP: CloudWatch at enterprise scale:

- **Cross-account observability**: centralise logs and metrics from all accounts into a monitoring account; uses CloudWatch cross-account sharing.
- **Anomaly Detection**: ML-based; auto-generates dynamic alarm bands from

metric history — no manual threshold needed.

- **Metrics Streams**: near-real-time stream of metrics to Kinesis Firehose → S3 / Splunk / Datadog. Alternative to polling the API.
- **ServiceLens + X-Ray**: distributed tracing integrated into CloudWatch console; trace request flow across Lambda, API Gateway, ECS, EC2.

31 AWS CloudTrail

SAA+SAP *API activity logging — who did what, when, from where*

[key] Key facts

- Records **every API call** to AWS services (management events by default).
- **Event types**: Management events (control plane — create/delete/modify), Data events (S3 object-level, Lambda invocations — paid extra), Insights events (unusual API patterns).
- Logs stored in **S3 bucket** (specify at trail creation). Enable **Log File Validation** to detect tampering (SHA-256 digest files).
- **Multi-region trail**: single trail covers all regions; logs to one bucket.
- **Retention**: CloudTrail event history (console): 90 days free. S3: indefinite.
- **Integration**: EventBridge can respond to CloudTrail events in near-real-time (e.g. alert on `DeleteBucket` or `StopInstances`).

[!] Exam tip

"Who deleted the S3 bucket?" ⇒ CloudTrail. **"Was there unusual API activity?"** ⇒ CloudTrail Insights. **"Detect crypto mining on EC2?"** ⇒ GuardDuty (analyses CloudTrail + VPC Flow Logs with ML).

32 AWS CloudFormation

SAA+SAP *Infrastructure as Code — provision resources via templates*

[key] Key facts

- Templates: JSON or YAML files defining AWS resources.
- **Stacks**: deployed instance of a template; resources are managed together.
- **Template sections**: AWSTemplateFormatVersion, Description, Parameters, Mappings, Conditions, Resources (required), Outputs.
- **Change Sets**: preview changes before applying (like a diff for infrastructure).
- **Stack Policies**: protect specific resources from being replaced/deleted.
- **Drift detection**: identify stack resources whose config has changed outside CloudFormation.
- **StackSets**: deploy stacks to multiple accounts and/or regions simultaneously.

[!] Exam tip

"Repeatable infrastructure deployments" or **"deploy same infra to multiple accounts"** ⇒ **CloudFormation StackSets**. **"Preview changes before deployment"** ⇒ **Change Sets**. CloudFormation uses **IAM service role** — the CF execution role must have permission to create all resources in the template.

[*] SAP depth

SAP: CloudFormation advanced:

- **Nested Stacks**: modularise large templates; child stack output becomes parent stack parameter. Reuse common patterns (network, IAM, compute layers).
- **Custom Resources**: invoke Lambda to provision resources not natively supported by CloudFormation (3rd-party APIs, on-prem systems).
- **Service Catalog**: wrap CloudFormation templates as Products; end users (developers) self-service provision approved products without direct CF access.
- **AWS CDK**: define infrastructure in TypeScript/Python/Java; CDK synthesises CloudFormation templates. Higher abstraction level; enables unit testing of IaC.
- **StackSets + Service-managed permissions**: use Organizations to automatically deploy to new accounts when they join the OU.

33 AWS Trusted Advisor

SAA *Best-practice checks across 5 categories*

[key] Key facts

- Inspects your AWS environment against best practices in 5 pillars: **Cost Optimisation**, **Performance**, **Security**, **Fault Tolerance**, **Service Limits**.
- **Basic/Developer plan**: 7 core checks (security + service limits).
- **Business/Enterprise plan**: all checks; programmatic access via API; CloudWatch integration for alerts.
- Security checks: open S3 buckets, exposed access keys, MFA on root, security group with unrestricted SSH, exposed RDS snapshots.
- **Service Limit checks**: warn when approaching account quotas.

34 AWS Config

SAP *Continuous resource configuration tracking and compliance*

[key] Key facts

- Records **configuration history** of AWS resources (who changed what, when, and to what). Unlike CloudTrail (API calls), Config tracks the *state* of resources over time.
- **Config Rules**: evaluate resources against desired configuration. Managed rules

(AWS-provided) or Custom rules (Lambda-backed).

- **Evaluation triggers**: on configuration change, periodic, or on-demand.
- **Conformance Packs**: bundle of Config rules + remediation actions as a deployable unit. Use for compliance standards (CIS, NIST, PCI-DSS, HIPAA).
- **Aggregator**: collect Config data from multiple accounts and regions into a central account for an org-wide compliance view.
- **Remediation**: auto-remediation via SSM Automation runbooks on rule violation (e.g. auto-enable encryption on non-compliant S3 bucket).

[!] Exam tip

Config vs CloudTrail:

- **CloudTrail** = "Who made the API call?" (who deleted the security group rule, when)
- **Config** = "What does the resource look like now vs before?" (was the S3 bucket public? is encryption enabled?)
- They are complementary: Config detects drift; CloudTrail identifies who caused it.

34.1 Continuous Compliance Services

Compliance and governance tools

Service	Purpose
AWS Config	Resource config tracking; compliance rules; drift detection; auto-remediation; org-wide aggregation
AWS Audit Manager	Automate evidence collection for compliance audits (PCI-DSS, HIPAA, SOC2, GDPR, ISO 27001). Maps AWS usage to audit controls continuously
AWS Artifact	On-demand access to AWS compliance reports (SOC, PCI, ISO) and agreements (BAA, NDA). No active scanning — document repository
Security Hub	Aggregates findings from Config, GuardDuty, Inspector, Macie, IAM Access Analyzer. CSPM (Cloud Security Posture Management). Single dashboard
AWS Firewall Manager	Centrally manage WAF, Shield Advanced, Security Groups, Network Firewall, Route 53 DNS Firewall across all accounts in an Organisation

[!] Exam tip

"Prove compliance to auditors" ⇒ **Audit Manager** (continuous evidence collection) + **Artifact** (AWS-issued compliance reports). **"Continuously detect non-compliant resources"** ⇒ **Config rules**. **"Central security dashboard across accounts"** ⇒ **Security Hub**. **"Apply WAF rules to all accounts automatically"** ⇒ **Firewall Manager**.

34.2 Patch Management Strategy

- **SSM Patch Manager**: define **Patch Baselines** (which patches are auto-approved by OS/severity/age); schedule patching via **Maintenance Windows** (low-traffic periods).
- **Amazon Inspector**: continuously scan EC2 and ECS container images for unpatched CVEs. Generates prioritised findings (critical/high/medium).
- **Immutable patching**: replace instances (launch new AMI) rather than patching in-place. Safer, predictable, no configuration drift. Use with Auto Scaling and golden AMIs.

35 AWS X-Ray

SAP *Distributed tracing for microservices and serverless*

[key] Key facts

- Traces requests as they travel through distributed application components.
- **Segments**: data from a single service (Lambda, EC2, ECS, API Gateway).
- **Subsegments**: finer-grained detail within a segment (DB call, HTTP request).
- **Traces**: end-to-end request journey across all segments.
- **Service Map**: visual graph of components and their connections; latency and error rates per edge.
- **X-Ray SDK**: instrument application code (Node.js, Python, Java, Go, Ruby, .NET).
- **X-Ray Daemon**: collects raw segment data and sends to X-Ray API (runs as a sidecar or agent on EC2/ECS).

Integrations (built-in, no SDK required):

- **Lambda**: enable Active Tracing in function configuration.
- **API Gateway**: enable X-Ray tracing on a stage.
- **ECS**: run X-Ray daemon as a sidecar container in the task definition.
- **Elastic Beanstalk**: enable via Beanstalk configuration option.

[!] Exam tip

X-Ray = **"find the slow part"** or **"trace errors across microservices"**. CloudWatch = **"aggregate metrics and alert"**. CloudTrail = **"audit who did what"**. All three together provide full observability: metrics (CloudWatch) + traces (X-Ray) + logs (CloudWatch Logs) + audit (CloudTrail).

[*] SAP depth

SAP: Observability stack:

- **CloudWatch ServiceLens**: combines CloudWatch metrics, CloudWatch Logs, and X-Ray traces in a unified console. Service map with health indicators.
- **Amazon Managed Grafana**: visualise metrics from CloudWatch, X-Ray, Prometheus, OpenSearch, and other data sources. Multi-account dashboards.
- **Amazon Managed Service for Prometheus**: open-source Prometheus-compatible monitoring for Kubernetes (EKS). Pairs with Managed Grafana.
- **Performance bottleneck decision framework:**
  - CPU-bound ⇒ scale out (more instances) or Compute Optimized family
  - Memory-bound ⇒ Memory Optimized instances; add ElastiCache
  - I/O-bound ⇒ upgrade EBS (gp3 → io2); use Instance Store; add Read Replicas
  - Network-bound ⇒ Enhanced Networking (SR-IOV); Placement Groups; check ENI limits
  - Database ⇒ Read Replicas; DAX/ElastiCache; query optimisation; connection pooling (RDS Proxy)

Part XII

Architecture Patterns

36 Key Architecture Patterns

SAA+SAP *Reference blueprints for the most common exam scenario architectures.*

These patterns appear repeatedly across both exams. For each one, know the services involved, why each service is chosen, and what the exam alternatives look like when constraints change (e.g. lowest cost, lowest RTO, no containers).

36.1 Three-Tier Web Application

[key] Key facts

- **Presentation tier**: Route 53 (DNS) → CloudFront (CDN/cache) → ALB (Layer-7 routing, SSL termination).
- **Application tier**: Auto Scaling Group of EC2 in *private subnets* across ≥2 AZs; or ECS Fargate tasks (no cluster management).
- **Data tier**: RDS Multi-AZ (primary write) + Read Replicas (read scaling) + ElastiCache Redis (session/object cache to reduce DB load).
- Static assets (HTML/CSS/JS/images) served from **S3 via CloudFront with OAC** — eliminates direct S3 public access.
- Use **S3 Gateway Endpoint** so EC2 instances reach S3 without traversing the internet (no NAT Gateway cost, no data-transfer charge).

[!] When the question changes the constraint

- **"Reduce DB read load"** → add ElastiCache Redis (cache-aside) or RDS Read Replica.
- **"Higher availability across regions"** → add Route 53 Failover + warm standby in second region; upgrade DB to Aurora Global Database.
- **"Least operational overhead"** → replace EC2 ASG with ECS Fargate; replace RDS with Aurora Serverless v2.
- **"Static IP required"** → replace ALB with NLB; add Elastic IP.

36.2 Serverless Web Application

[key] Key facts

- **Full stack**: Route 53 → CloudFront → API Gateway → Lambda → DynamoDB (+ DAX for sub-ms reads).
- **Static content**: S3 bucket serving the SPA/frontend via CloudFront with OAC.
- **Auth**: Amazon Cognito User Pools (user directory + JWT tokens) + Cognito Identity Pools (exchange token for temporary AWS credentials).
- **No servers to patch**; scales automatically to zero and to millions; pay per request only.
- Lambda@Edge or CloudFront Functions can handle auth, redirects, and header manipulation at the edge — before the request reaches API Gateway.

Serverless web app — service substitutions

Need	Service
HTTP API, low latency, cheap	API Gateway HTTP API (not REST API; 70% cheaper)
REST API with full features (throttle, cache, keys)	API Gateway REST API
Simple web app / API, full managed	AWS App Runner (from container or source)
Background async processing	Lambda triggered by SQS / EventBridge
Long-running tasks (>15 min)	ECS Fargate task (replace Lambda)
Complex multi-step workflow	AWS Step Functions

36.3 Event-Driven / Decoupled Pipeline

[key] Key facts

- **Pattern**: Producer → SQS / Kinesis Data Streams → Lambda / ECS workers → S3 / DynamoDB.
- **Fan-out**: SNS topic → multiple SQS queues; each queue drives an independent downstream Lambda or worker — allows parallel processing without coordination.
- **Rule-based routing**: EventBridge event bus; pattern rules filter events to specific Lambda/SQS/Step Functions targets; preferred over SNS for event-driven architectures reacting to AWS service events or SaaS.
- **Failure handling**: Dead-letter queue (DLQ) on SQS or Lambda; messages that exceed `maxReceiveCount` land in DLQ for inspection.
- **Ordered + replay**: Kinesis Data Streams (up to 7-day replay, ordered per shard); use when consumers need to replay or process in order.

[!] SQS vs Kinesis decision

- **SQS**: decouple, buffer, unknown order OK, each message processed once → **SQS Standard**.
- **SQS FIFO**: strict order, exactly-once processing (financial transactions, inventory deduction).
- **Kinesis Data Streams**: ordered per shard, replay up to 7 days, multiple consumers reading concurrently (fan-out without copying messages).
- **Kinesis Firehose**: fully managed delivery to S3/Redshift/OpenSearch; no consumer code needed; near-real-time (60 s buffer or 1 MB).

36.4 Data Lake Architecture

[key] Key facts

- **Ingestion**: Kinesis Data Firehose (streaming) or AWS DataSync (bulk online) or Snow Family (offline) → S3 raw/landing zone.
- **Catalogue & ETL**: AWS Glue (serverless ETL + Data Catalog) transforms raw data → S3 curated/processed zone in Parquet/ORC.
- **Query**: Amazon Athena (SQL on S3; pay per query; no infrastructure) or Redshift Spectrum (query S3 directly from a Redshift cluster).
- **Visualise**: Amazon QuickSight (BI dashboards; connects to Athena/Redshift/S3).
- **Access control**: S3 bucket policies + Lake Formation (column/row-level security on the data catalog).

[!] Data lake query pattern shortcuts

- **"Ad-hoc queries on S3 without loading" → Athena.**
- **"Join S3 data with Redshift warehouse" → Redshift Spectrum.**
- **"Centralise metadata and permissions across all data" → AWS Lake Formation.**
- **"Serverless ETL with dependency tracking" → AWS Glue.**

36.5 Hybrid Cloud Pattern

[key] Key facts

- **Network**: on-prem ↔ Direct Connect (consistent, private, lower latency) *or* Site-to-Site VPN (lower cost, internet, encrypted). Use **both** for redundancy (VPN as failover for DX).
- **File storage extension**: Storage Gateway (File mode) → S3; local cache keeps hot files fast; cloud stores everything.
- **Automated data transfer**: AWS DataSync (scheduled/incremental; NFS/SMB/HDFS → S3/EFS/FSx); up to 10 Gbps per task.
- **Bulk offline transfer**: Snowball Edge (terabytes; 80 TB per device) or Snowmobile (petabytes; 100 PB per truck) when bandwidth is insufficient.
- **DNS**: Route 53 Resolver inbound/outbound endpoints allow private DNS resolution between on-prem DNS and Route 53 Private Hosted Zones.
- **Identity**: AWS Directory Service (Managed Microsoft AD) for SSO between on-prem Active Directory and AWS resources.

Direct Connect vs VPN

	Direct Connect	Site-to-Site VPN
Path	Dedicated private fibre	Encrypted over internet
Latency	Consistent, low	Variable
Bandwidth	1 Gbps – 100 Gbps	Up to 1.25 Gbps
Cost	Higher (port + data)	Lower
Setup time	Weeks	Minutes
Encryption	Not by default (add VPN on DX)	Always encrypted (IPsec)
Best for	Steady high-volume, compliance	Backup/failover, quick setup

36.6 Migration Services Reference

Migration tool selection

Service	Use case
AWS DMS (Database Migration Service)	Migrate relational/NoSQL DBs; heterogeneous migrations; CDC for minimal downtime
AWS SCT (Schema Conversion Tool)	Convert schema from one engine to another (Oracle → Aurora); runs alongside DMS
AWS DataSync	Online automated transfer; NFS/SMB/HDFS → S3/EFS/FSx; up to 10 Gbps
AWS Snowball Edge	Offline bulk transfer; 80 TB usable storage; compute at edge option
AWS Snowmobile	Exabyte-scale (≥100 PB); physical truck transport
AWS Application Migration Service (MGN)	Lift-and-shift; continuous block-level replication; minimal-downtime cutover
AWS Transfer Family	Managed SFTP/FTPS/FTP/AS2 to S3 or EFS; drop-in replacement for on-prem FTP servers

[!] Migration exam shortcuts

- **"Transfer 2 PB with limited bandwidth"** → **Snowball Edge** (multiple devices).
- **"Transfer 100 PB or more"** → **Snowmobile**.
- **"Ongoing DB migration, minimal downtime"** → **DMS with CDC** (Change Data Capture).
- **"Oracle to Aurora"** → **SCT** (schema) + **DMS** (data).
- **"Lift-and-shift 200 servers"** → **Application Migration Service (MGN)**.

- **"Automated periodic transfer from on-prem NAS to S3" → DataSync.**

[*] SAP depth

SAP migration depth: SAP-C02 adds the **7Rs framework** (Retire, Retain, Rehost, Relocate, Replatform, Repurchase, Refactor) as a *decision methodology*, not just a tool list. Refactor (re-architect to cloud-native) has the highest long-term benefit but the highest effort. Rehost (lift-and-shift via MGN) has the lowest benefit but lowest effort and is the fastest path. SAP scenarios will ask you to recommend the right "R" for a given business constraint.

Part XIII

Quick Reference

37 Quick Reference & Decision Trees

SAA+SAP *Critical numbers, service selection rules, and exam-day strategy.*

37.1 Critical Numbers to Memorise

Limits and numbers the exam loves to test

Item	Value
S3 single PUT max size	5 GB
S3 object max size (multipart)	5 TB
S3 durability	99.999 999 999% (11 nines)
S3 Standard availability	99.99%
SQS max message size	256 KB
SQS max retention period	14 days
SQS visibility timeout max	12 hours
Lambda max execution time	15 minutes
Lambda max memory	10,240 MB (10 GB)
Lambda max deployment (zip)	50 MB (250 MB unzipped)
API Gateway max payload	10 MB
DynamoDB item max size	400 KB
EBS gp3 max volume	16 TiB
EBS io2 Block Express max IOPS	256,000 IOPS
RDS max storage	64 TiB (gp2/gp3)
CloudFront max file size	20 GB
Route 53 TTL minimum	1 second
Multipart upload threshold	Recommended above 100 MB; required above 5 GB

37.2 Service Selection Decision Trees

37.2.1 Which Messaging / Queue Service?

[key] Key facts

- Decouple application tiers, buffer bursts → **SQS Standard**
- Ordered messages, no duplicates (financial transactions) → **SQS FIFO**
- Fan-out to multiple consumers simultaneously → **SNS**
- Fan-out to multiple SQS queues (each processes independently) → **SNS** → **SQS**
- Rule-based routing, SaaS integration, event replay → **EventBridge**
- Migrate existing ActiveMQ / RabbitMQ → **Amazon MQ**
- Real-time ordered streaming, replay, multiple consumers → **Kinesis Data Streams**
- Complex workflow with retries, error handling, branching → **Step Functions**

37.2.2 Which Load Balancer?

[key] Key facts

- HTTP/HTTPS; path-based or host-based routing; WebSocket → **ALB**
- TCP/UDP; static IP required; ultra-low latency; gaming → **NLB**
- Third-party network appliances (firewalls, IDS); bump-in-the-wire → **GLB**
- Legacy (EC2-Classic); avoid for new projects → **CLB**

37.2.3 Which Storage Service?

[key] Key facts

- Objects, backups, web assets, user uploads → **S3**
- OS disk, database data volume (single EC2) → **EBS**
- Shared POSIX file system across Linux EC2 instances → **EFS**
- Shared SMB file system for Windows workloads → **FSx for Windows File Server**
- HPC / ML high-throughput parallel FS → **FSx for Lustre**
- Lift-and-shift NetApp (NFS, SMB, iSCSI) → **FSx for NetApp ONTAP**
- Archive / cold, infrequent restore (hours) → **S3 Glacier Deep Archive**
- Offline bulk data transfer (terabytes/petabytes) → **Snow Family**
- Hybrid: on-prem accessing AWS storage → **AWS Storage Gateway**

37.2.4 Which Database?

[key] Key facts

- Relational, OLTP, SQL, complex queries → **RDS / Aurora**
- Global, multi-region relational with near-zero RPO/RTO → **Aurora Global Database**
- Key-value / document, any scale, millisecond latency → **DynamoDB**
- Multi-region active-active K/V → **DynamoDB Global Tables**
- OLAP, data warehousing, petabyte analytics → **Redshift**
- In-memory sessions, caching, leaderboards → **ElastiCache Redis**
- Graph (social networks, fraud detection) → **Neptune**
- Search, log analytics, full-text search → **OpenSearch Service**
- MongoDB-compatible document store → **DocumentDB**
- IoT / time-series metrics → **Timestream**
- Durable in-memory Redis-compatible → **MemoryDB for Redis**

37.2.5 CloudFront vs Global Accelerator

CloudFront vs Global Accelerator

	CloudFront	Global Accelerator
Protocol	HTTP/HTTPS (Layer 7)	TCP/UDP (Layer 4)
Caching	Yes — content cached at edge	No caching
Best for	Static/dynamic web content, CDN	Non-HTTP apps, gaming, IoT, consistent performance
IP type	Dynamic (DNS)	Static Anycast IP
Failover	Origin failover	30-second endpoint failover

37.3 Exam-Day Strategy

[!] SAP-C02 Exam Tips

1. **Read the question twice** — SAP questions are long; identify what is *actually* being asked (RTO? cost? operational overhead?)
2. **Look for constraints** — RTO/RPO numbers, budget limits, existing investments, compliance requirements
3. **Eliminate obviously wrong answers first** — usually 1–2 can be ruled out immediately (single AZ, storing creds on EC2, etc.)
4. **Well-Architected lens** — when in doubt, ask which answer best aligns with AWS best practices
5. **Managed > Self-managed** — AWS exam answers favour managed services

over DIY on EC2

6. **Multi-AZ is table stakes** — any answer placing a critical workload in a single AZ is almost certainly wrong
7. **IAM roles over access keys** — always prefer roles for AWS-to-AWS access
8. **"Least operational overhead"** — often points to serverless or fully managed services (Fargate over EC2, Aurora over RDS on EC2)
9. **"Most cost-effective"** — changes the answer; Spot for fault-tolerant batch, Savings Plans for steady-state, S3 Intelligent-Tiering for unknown access patterns
10. **Time management** — approximately 2.5 minutes per question; flag hard ones and return

37.4 Categorised Service Cheat Sheet

37.4.1 Compute

[key] Key facts

- **EC2 On-Demand** — short-term, unpredictable; development/test; never choose for cost-optimisation questions.
- **EC2 Reserved / Savings Plans** — steady-state, predictable workloads (1 or 3 year); up to 72% savings.
- **Spot Instances** — fault-tolerant batch, flexible start/stop; up to 90% savings; 2-min interruption warning; never for prod DBs.
- **Dedicated Host** — BYOL licensing; physical server isolation; compliance requirements.
- **Lambda** — event-driven, short duration ($\leq$15 min), no idle cost; pairs with SQS/SNS/API Gateway.
- **Fargate** — serverless containers; no cluster management; pay per task; variable/bursty loads.
- **Auto Scaling** — Target Tracking = simplest; Step = fine-grained; Scheduled = known patterns; Predictive = ML-based.
- **Elastic Beanstalk** — PaaS; developer uploads code; AWS manages EC2/ALB/ASG/RDS; minimal overhead.

37.4.2 Storage

[key] Key facts

- **S3 Standard** — frequent access; 11 nines durability; most expensive storage class.
- **S3 Standard-IA** — infrequent access; same durability; lower storage cost, per-retrieval fee.
- **S3 Intelligent-Tiering** — unknown access patterns; automatic tiering; no retrieval fee for monitored tier.
- **S3 Glacier Instant Retrieval** — archive accessed once a quarter; millisecond retrieval.

- **S3 Glacier Flexible Retrieval** — minutes to hours retrieval; bulk cheapest.
- **S3 Glacier Deep Archive** — lowest cost; 12-hour standard retrieval; long-term compliance archive.
- **EBS gp3** — general purpose SSD; independent IOPS/throughput tuning; prefer over gp2.
- **EBS io2** — highest IOPS (256K); SAP HANA; Multi-Attach (single AZ only).
- **EBS st1/sc1** — HDD; sequential throughput (st1) or cold/infrequent (sc1); *never* for boot volumes.
- **Instance Store** — ephemeral; highest I/O; data lost on stop/terminate; use for temp data, buffers.
- **EFS** — NFS; Linux only; multi-AZ; elastic capacity; shared across many EC2 instances/Lambda.
- **FSx for Windows** — SMB; Active Directory integration; Windows workloads.
- **FSx for Lustre** — HPC/ML; sub-millisecond; integrates with S3 for burst-out data lake training.
- **Storage Gateway File mode** — NFS/SMB on-prem; objects land in S3; good for gradual migration.

37.4.3 Databases

[key] Key facts

- **RDS Multi-AZ** — synchronous standby; automatic failover (∼60–120 s); NOT a read replica.
- **RDS Read Replicas** — asynchronous; up to 15 replicas (Aurora); used for read scaling, not HA.
- **Aurora** — 5× MySQL / 3× PostgreSQL performance; 6-way replication across 3 AZs; storage auto-scales.
- **Aurora Global Database** — 1 primary + up to 5 secondary regions; RPO <1 s; RTO <1 min; cross-region DR.
- **Aurora Serverless v2** — auto-scales capacity in fractions of an ACU; ideal for variable/dev workloads.
- **DynamoDB** — key-value/document; single-digit ms at any scale; serverless; TTL; streams.
- **DynamoDB Global Tables** — multi-region active-active; conflict resolution = last-write-wins.
- **DAX** — DynamoDB Accelerator; in-memory cache; microsecond reads; drop-in (just change endpoint).
- **ElastiCache Redis** — sub-ms; persistence; pub/sub; sorted sets; sessions; leaderboards.
- **Redshift** — columnar OLAP; petabyte-scale; Redshift Spectrum queries S3 without loading.
- **Neptune** — graph; social networks, fraud detection, recommendation engines.
- **Timestream** — time-series; IoT metrics; built-in analytics.

37.4.4 Networking

[key] Key facts

- **ALB** — Layer 7; path-based / host-based / header routing; WebSocket; Lambda targets; WAF integration.
- **NLB** — Layer 4 TCP/UDP; static IP (Elastic IP); ultra-low latency; TLS termination; PrivateLink.
- **GWLB** — Layer 3; bump-in-the-wire for 3rd-party virtual appliances (firewalls, IDS/IPS).
- **CloudFront** — CDN; Layer 7 caching; OAC for S3 private content; Lambda@Edge / CloudFront Functions at edge.
- **Global Accelerator** — static Anycast IPs; Layer 4; no caching; consistent global performance; 30 s failover.
- **Route 53 Failover** — active-passive; health checks required; use with secondary S3 static site.
- **Route 53 Latency** — routes to region with lowest measured latency for the user.
- **Route 53 Geolocation** — routes based on where the user is; default record for no match.
- **Route 53 Weighted** — A/B testing; gradual traffic shifting; set weight 0 to stop traffic.
- **VPC Gateway Endpoint** — free; route S3 and DynamoDB traffic within VPC; eliminates NAT Gateway charges.
- **VPC Interface Endpoint (PrivateLink)** — private connectivity to other AWS services / SaaS; ENI in subnet.
- **Transit Gateway** — hub-and-spoke; connect many VPCs + on-prem; shared across accounts.
- **Direct Connect** — dedicated private connection; consistent bandwidth; lower egress pricing; use with VPN for redundancy.

37.4.5 Security

[key] Key facts

- **IAM Roles** — preferred over long-term access keys; use for EC2, Lambda, cross-account, federation.
- **KMS** — envelope encryption; CMK never leaves KMS; use `GenerateDataKey` for client-side encryption.
- **SSE-S3** — AWS manages keys; free; least admin overhead for S3 encryption.
- **SSE-KMS** — customer controls key policy; audit via CloudTrail; required for compliance.
- **SSE-C** — customer provides key per request; AWS does not store the key.
- **Secrets Manager** — auto-rotation; integrates with RDS/Redshift/DocumentDB natively; prefer for DB creds.
- **Parameter Store** — free tier (Standard); store config, non-sensitive params;

SecureString uses KMS.

- **GuardDuty** — threat detection using ML; analyses VPC Flow Logs, DNS, CloudTrail; no agents needed.
- **Inspector** — vulnerability scanning; EC2 and container images; CVE/CIS benchmarks; integrates with ECR.
- **Macie** — PII/sensitive data discovery in S3; ML-based; privacy compliance.
- **Shield Standard** — free; automatic L3/L4 DDoS protection for all AWS customers.
- **Shield Advanced** — paid; L7 protection; WAF integration; DDoS cost protection; 24/7 SRT access.
- **WAF** — web ACL; L7 rules (IP, geo, rate, SQL injection, XSS); attaches to ALB/CloudFront/API GW.
- **Security Groups** — stateful; instance level; allow rules only.
- **NACLs** — stateless; subnet level; allow and deny rules; numbered rule order.

37.4.6 High Availability & Disaster Recovery

[key] Key facts

- **Multi-AZ** — synchronous replication; automatic failover; HA within a region; not a read-scaling solution.
- **Multi-Region** — active-passive (failover) or active-active (Global Tables, Aurora Global); cross-region DR.
- **Pilot Light** — minimal version always running; scale up on disaster; RTO: tens of minutes.
- **Warm Standby** — reduced-capacity version always running; scale to full on disaster; RTO: minutes.
- **Multi-site Active/Active** — full capacity in both regions; near-zero RTO/RPO; highest cost.
- **Backup & Restore** — cheapest; highest RTO/RPO (hours); restore from S3/Glacier; fine for non-critical.
- **S3 CRR** — cross-region replication; requires versioning on both buckets; not retroactive; same-account or cross-account.
- **RDS Automated Backups** — daily snapshot + transaction logs; restore to any point in retention window (up to 35 days).
- **RDS Snapshot** — manual; not subject to retention period; can copy across regions for DR.
- **SQS as buffer** — decouple components; absorb traffic spikes; prevent cascade failures; standard (∞ scale) or FIFO (3K TPS).
- **Health checks** — Route 53 health checks trigger DNS failover; ALB health checks deregister unhealthy targets.

37.5 The Four Questions Framework

[!] Answering Strategy — ask these four questions

Before selecting an answer, mentally run through these four questions in order:

1. **What is the actual requirement?** Read the scenario once for the story, then once more to extract the *hard constraint*: is it RTO, RPO, cost, latency, compliance, or operational overhead?
2. **What can I eliminate immediately?** Look for classic wrong-answer patterns: single AZ for critical workloads, long-term access keys on EC2, On-Demand for cost questions, manual processes when automation is available, self-managed when a managed service exists.
3. **Which remaining answer is the most "AWS-native" solution?** AWS exam answers reward managed services, IAM roles over keys, encryption by default, and decoupled/serverless architectures. When two answers both seem correct, the more managed / less operationally complex one is usually right.
4. **Does it satisfy the constraint from question 1?** Double-check: if the question said "lowest cost," confirm your choice is not over-engineered. If it said "near-zero RTO," confirm it is not Backup & Restore.

37.6 High-Frequency Scenario Pattern Tables

37.6.1 "Without Internet" Patterns

Private connectivity scenarios

Scenario	Answer
EC2 must access S3 without internet	S3 Gateway VPC Endpoint (free)
EC2 must access DynamoDB without internet	DynamoDB Gateway VPC Endpoint (free)
Lambda / ECS in VPC needs S3 / DynamoDB	Gateway VPC Endpoint
Private subnet EC2 needs outbound internet	NAT Gateway in public subnet + route table entry
VPC needs private access to other AWS service (non S3/DDB)	Interface Endpoint (PrivateLink)
On-prem server needs to call an AWS service privately	Interface Endpoint via Direct Connect / VPN

37.6.2 HA & DR Patterns

High availability and disaster recovery

Scenario	Answer
Survive single AZ failure	Multi-AZ deployment (ALB + ASG + RDS Multi-AZ)
Fast automatic DB failover, same region	RDS Multi-AZ (∼60 s) or Aurora (<30 s)
DB readable from any global region	Aurora Global Database
Active-active key-value globally	DynamoDB Global Tables
Lowest RPO/RTO across regions (any app)	Active-active + Route 53 health checks
Cost-sensitive DR, hours RTO acceptable	Backup & Restore to S3 + CloudFormation redeploy
Minimal always-on cost DR (core DB only)	Pilot Light (scale out on event)
Faster failover but reduced capacity ready	Warm Standby (scale up on event)

37.6.3 Performance & Caching Patterns

Performance and caching

Scenario	Answer
Read-heavy relational DB; reduce load	ElastiCache Redis (cache-aside pattern)
DynamoDB hot partitions / high read latency	Enable DAX (same API, microsecond reads)
RDS read scaling	Read Replicas (up to 5 RDS, 15 Aurora)
Static content for global users	S3 + CloudFront (OAC)
Dynamic content globally with fixed IP	Global Accelerator (Anycast, Layer 4)
EC2 low-latency inter-node communication	Cluster Placement Group
Lambda cold starts in VPC	Provisioned Concurrency
Containers, no cluster management	ECS Fargate
HPC/ML storage, sub-ms parallel FS	FSx for Lustre (integrate with S3)

37.6.4 Cost Optimisation Patterns

Cost optimisation

Scenario	Answer
Steady-state predictable EC2 workload	Reserved Instances or Compute Savings Plans
Fault-tolerant batch / big data processing	Spot Instances (up to 90% savings)
Minimum 2 always-on + burst capacity	2 Reserved + Spot for burst
S3 data accessed infrequently but occasionally	S3 Standard-IA (retrieval fee applies)
S3 access pattern unknown / changing	S3 Intelligent-Tiering (auto-tier, no retrieval fee)
Long-term compliance archive (≥10 years)	S3 Glacier Deep Archive
Variable workload, no idle compute charge	Lambda / Aurora Serverless / DynamoDB On-Demand
Avoid cross-AZ data transfer charges	Use private IP addresses; same-region same-AZ
Large offline data transfer to AWS	Snowball Edge (<10 PB); Snowmobile (≥100 PB)

37.7 One-Liner Rapid-Fire Rules

37.7.1 Security

[key] Key facts

- IAM role on EC2/Lambda ⇒ **never** embed access keys in code or config.
- Explicit Deny **always wins** in IAM policy evaluation — no Allow overrides it.
- Access S3 from VPC ⇒ **Gateway endpoint**, not NAT Gateway.
- Auto-rotate database credentials ⇒ **Secrets Manager**, not Parameter Store.
- DDoS at Layer 7 ⇒ **WAF + Shield Advanced** on CloudFront or ALB.
- Encrypt existing unencrypted EBS/RDS ⇒ **snapshot → copy with encryption → restore**.
- Immutable S3 archive (no one can delete) ⇒ **Object Lock Compliance mode**.
- Enforce guardrails across all accounts ⇒ **SCPs via AWS Organizations**.
- Security Group = **stateful** (return traffic auto-allowed); NACL = **stateless** (both directions explicit).
- Cross-account access ⇒ IAM role + STS AssumeRole; never share credentials.

37.7.2 Resilience

[key] Key facts

- RDS Multi-AZ ⇒ **failover only**, not for read scaling.
- RDS Read Replica ⇒ **reads / reporting**, not HA failover.
- Aurora automatic failover <30 s; RDS Multi-AZ ≈60 s.
- NLB ⇒ static IP / TCP-UDP; ALB ⇒ path routing / WebSockets.
- SQS DLQ ⇒ handle failed / poison-pill messages after `maxReceiveCount`.
- Decouple and buffer traffic bursts ⇒ **SQS Standard**.
- Fan-out simultaneously to multiple services ⇒ **SNS** → **multiple SQS queues**.
- Strict order + exactly-once processing ⇒ **SQS FIFO**.
- Complex stateful workflow with retries ⇒ **Step Functions**.
- Route 53 failover requires a **health check** on the primary record.

37.7.3 Performance

[key] Key facts

- HPC inter-node low latency ⇒ **Cluster Placement Group** (same rack).
- gp3 = default EBS (tunable IOPS/throughput); io2 = highest IOPS (256K).
- DynamoDB sub-millisecond reads ⇒ **DAX** in front (drop-in, same API).
- ElastiCache **Redis** ⇒ sorted sets, pub/sub, persistence, HA, sessions.
- ElastiCache **Memcached** ⇒ simple K/V, multi-threaded, horizontal scale, no persistence.
- Lambda cold start in VPC ⇒ **Provisioned Concurrency** pre-warms execution environments.
- Containers, no cluster management ⇒ **ECS Fargate**; need Kubernetes ⇒ **EKS**.
- Global static/dynamic content ⇒ **CloudFront** + OAC.
- Non-HTTP global + fixed IP ⇒ **Global Accelerator** (Anycast IPs).
- Route 53 Latency routing ⇒ lowest measured latency for the end user.

37.7.4 Cost

[key] Key facts

- Steady workload ⇒ **Reserved Instances** (standard) or **Compute Savings Plans**.
- Fault-tolerant batch ⇒ **Spot Instances** (up to 90% off; 2-min warning).
- Unknown S3 access pattern ⇒ **S3 Intelligent-Tiering** (no retrieval fee).
- Long archive (≥180 days) ⇒ **S3 Glacier Deep Archive** (≈ $0.001/GB/month).
- No idle compute cost ⇒ **Lambda / DynamoDB On-Demand / Aurora Serverless v2**.
- Cross-AZ free ⇒ use **private IPs** within the same region (public/EIP = charged).
- Large offline transfer ⇒ **Snowball Edge** (<10 PB); **Snowmobile** (≥100 PB).

- DB migration minimal downtime ⇒ **DMS + CDC**.
- On-prem file server, capacity full ⇒ **Storage Gateway File** (NFS/SMB → S3).
- Compliance evidence ⇒ **CloudTrail** (API logs) + **Config** (resource config) + **GuardDuty** (threat detection).

37.8 Memory Hooks — One-Word Mnemonics

Single-word memory hooks for exam day

Service	Hook	What it does in one phrase
Amazon CloudWatch	OBSERVE	Metrics, alarms, logs, dashboards
AWS CloudTrail	AUDIT	API call history — who did what, when, from where
AWS Config	RECORD	Config changes and compliance state over time
Amazon GuardDuty	DETECT	ML threat detection on CloudTrail / Flow Logs / DNS
Amazon Inspector	FIND VULNS	CVEs in EC2, containers, Lambda
Amazon Macie	FIND PII	Sensitive data discovery in S3
AWS Security Hub	CENTRALIZE	Aggregates GuardDuty, Inspector, Macie findings
Amazon Detective	INVESTIGATE	Drill into GuardDuty events; visualise relationships
Amazon Athena	SQL on S3	Serverless query; pay per query; no ETL
AWS Glue	ETL	Serverless transform + Data Catalog
Amazon Redshift	WAREHOUSE	OLAP, columnar, petabyte-scale analytics
Amazon Kinesis DS	STREAM	Real-time ingest, ordered, replay
Amazon Data Firehose	DELIVER	Load streams to S3/Redshift/OpenSearch
Amazon CloudFront	CDN	Edge caching, HTTP/HTTPS content delivery
Global Accelerator	GLOBAL IP	TCP/UDP, fixed anycast IPs, non-HTTP global
Transit Gateway	HUB ROUTER	Connects many VPCs + on-prem
AWS Direct Connect	PRIVATE LINK	Dedicated line to AWS; not over internet
AWS X-Ray	TRACE	Distributed tracing; find performance bottlenecks

Part XIV

Practice Questions

38 SAP-C02 Practice Questions

These questions are SAP-C02 level. Multiple-response questions specify how many answers to select. Explanations cover why the correct answer is right AND why the distractors are wrong.

38.1 Domain 1 — Design for Organizational Complexity

Question 1 SAP (multi-account logging)

A company has 150 AWS accounts managed under AWS Organizations. The security team requires that all API activity across every account is logged immutably in a centralised location and that no individual account administrator can delete or modify those logs. Which combination of actions meets these requirements with the *least operational overhead*? (Select TWO.)

A. Enable AWS CloudTrail in each account individually and configure an S3 bucket in each account to store logs.

B. Create an organisation-level CloudTrail trail from the management account that logs to an S3 bucket in a dedicated Log Archive account.

C. Apply an S3 bucket policy on the Log Archive bucket that allows `s3:PutObject` from all organisation accounts but explicitly denies `s3:DeleteObject` and `s3:DeleteBucket`.

D. Use AWS Config with an aggregator to forward CloudTrail logs from each account to the central bucket.

E. Attach an SCP to the root OU that denies `cloudtrail:DeleteTrail` and `cloudtrail:StopLogging` for all accounts except the management account.

Correct Answer: B, E

B is correct: an organisation-level CloudTrail trail created from the management account automatically applies to all current and future member accounts and centralises logs to a single S3 bucket — minimal per-account setup needed.
E is correct: an SCP applied at the Root OU that denies `cloudtrail:DeleteTrail` and `cloudtrail:StopLogging` prevents any member account administrator from disabling the trail. SCPs override even account-level admin permissions.
A is wrong: per-account trails are high overhead (150 separate configurations) and do not prevent deletion in each account.
C would protect the S3 bucket but does not prevent someone from disabling the trail at the source account.
D is wrong: AWS Config is a configuration compliance service, not a mechanism for forwarding CloudTrail log files.
Key concept: combine the organisation CloudTrail trail (B) with SCPs that prevent disabling it (E) for end-to-end immutability.

Question 2 SAP (SCPs and IAM policy evaluation)

An AWS Organizations management account administrator attaches a Service Control Policy (SCP) to the Production OU with an explicit `Allow` for `s3:*`. A developer in an account under that OU has an IAM policy that allows all `s3:*` actions. The developer attempts to create an S3 bucket and receives an `Access Denied` error. What is the MOST likely cause?

A. The SCP `Allow` is overriding the IAM policy.

B. The management account requires a separate IAM role for member accounts.

C. There is an explicit `Deny` SCP at a higher level in the OU hierarchy (e.g. the Root OU) that overrides the `Allow` SCP.

D. SCPs do not apply to S3 actions.

Correct Answer: C

C is correct: SCP evaluation is hierarchical. If a parent OU (or the Root) has an explicit `Deny` for `s3:CreateBucket`, it overrides any `Allow` at a lower level. Also remember: SCPs do NOT grant permissions — an SCP `Allow` alone is insufficient; both the SCP and the IAM policy must allow the action, but an SCP `Deny` at any level blocks the action regardless.
A is wrong: SCP Allows do not override IAM; they set the maximum boundary.
B is wrong: cross-account access is handled by IAM roles, but within an account the developer's own IAM policy applies.
D is wrong: SCPs apply to all AWS service actions.
Key insight: explicit Deny always wins. Check the full SCP inheritance chain when debugging access denied errors in a multi-account org.

Question 3 SAP (VPC connectivity)

A company has 30 VPCs across three AWS regions. Each VPC needs to communicate with every other VPC, and all VPCs need to connect to an on-premises data centre via Direct Connect. The team must minimise operational overhead. Which architecture achieves this?

A. Create VPC peering connections in a full mesh between all 30 VPCs and use a separate Site-to-Site VPN to each VPC for on-premises connectivity.

B. Deploy one Transit Gateway (TGW) per region, peer the regional TGWs together, attach all VPCs to their regional TGW, and connect to on-premises via a Direct Connect Gateway attached to each regional TGW.

C. Use AWS PrivateLink to connect each VPC and use Direct Connect for on-premises.

D. Deploy one VPC as a hub, peer all other VPCs to it, and use that hub for on-premises connectivity.

Correct Answer: B

B is correct: Transit Gateway per region, peered across regions, with a Direct Connect Gateway providing on-premises connectivity through transit virtual interfaces — this is the standard AWS recommended multi-region hub-and-spoke architecture. It

scales to hundreds of VPCs with low operational overhead.
A is wrong: a full mesh of 30 VPCs requires $30 \times 29/2 = 435$ peering connections and is operationally unmanageable. VPC peering is also not transitive, so it does not simplify on-premises routing.
C is wrong: PrivateLink is for one-directional service exposure (provider → consumer), not general any-to-any VPC routing.
D is wrong: the "hub" VPC with peering is not transitive — spokes cannot communicate with each other via the hub.

38.2 Domain 2 — Design for New Solutions

Question 4 SAP (disaster recovery strategy selection)

A financial services company runs a critical OLTP application on RDS Aurora MySQL. The business requires an RPO of 5 seconds and an RTO of 1 minute in the event of a regional failure. Cost must be minimised while still meeting these requirements. Which solution meets the requirements?

A. Configure RDS Multi-AZ with automated backups to S3 Cross-Region Replication. On failure, restore from the S3 backup in the DR region.
B. Use Amazon RDS with a cross-region read replica. On failure, manually promote the read replica and update DNS.
C. Configure Aurora Global Database with the primary in `us-east-1` and a secondary in `ap-southeast-1`. Enable Aurora's Fast Database Cloning for failover.
D. Configure Aurora Global Database with the primary in `us-east-1` and a secondary cluster in `ap-southeast-1`. On failure, perform a managed planned failover or unplanned failover to promote the secondary.

Correct Answer: D

D is correct: Aurora Global Database achieves an RPO of less than 1 second (continuous asynchronous replication at the storage layer) and an RTO of less than 1 minute (promote the secondary cluster). This directly meets the 5-second RPO and 1-minute RTO requirements.
A is wrong: S3 CRR + restore from backup has an RTO of hours — far exceeding the 1-minute requirement.
B is wrong: RDS cross-region read replicas use async replication with RPO of potentially minutes; manual promotion adds to RTO and does not guarantee sub-5-second RPO.
C is wrong: Fast Database Cloning is for creating test copies, not DR failover.
Key numbers: Aurora Global — RPO < 1 s, RTO < 1 min. Use this any time the exam specifies sub-second RPO or sub-minute RTO for a relational database.

Question 5 SAP (deployment strategy)

A company needs to deploy a new version of a web application running on Amazon ECS (behind an ALB) with zero downtime and the ability to *instantly rollback* if error rates increase. Which deployment strategy should they use?

A. Rolling update: gradually replace old task definitions with new ones in the existing ECS service.
B. Blue/Green deployment using AWS CodeDeploy: deploy the new version as a separate target group and shift traffic via ALB listener rules; rollback by re-weighting to the old target group.
C. All-at-once deployment: stop all old tasks, then start all new tasks simultaneously.
D. Canary deployment via Route 53 weighted routing: shift 10% of traffic to the new version and monitor before shifting 100%.

Correct Answer: B

B is correct: Blue/Green with CodeDeploy and ALB target group weights allows instant traffic shifting (or canary
A is wrong: rolling updates do not have instant rollback — you must re-deploy the old version through another rolling cycle.
C is wrong: all-at-once means downtime while tasks restart.
D is valid for gradual traffic shifting but Route 53 routing operates at DNS level with TTL delays, making rollback slower than ALB target group re-weighting. Also, CodeDeploy can do canary and linear deployment natively for ECS, making it a better integrated solution.

38.3 Domain 3 — Continuous Improvement

Question 6 SAP (performance monitoring)

A company's three-tier web application shows intermittent high latency that only affects specific API endpoints. The team needs to identify *which microservice* in the call chain is responsible and what the downstream dependencies look like. Which service should they use?

A. Amazon CloudWatch Metrics with custom dashboards
B. Amazon CloudWatch Log Insights with query filters
C. AWS X-Ray distributed tracing
D. Amazon Inspector

Correct Answer: C

C is correct: AWS X-Ray provides distributed tracing — it instruments the application to track requests as they flow through microservices, generating a *service map* that shows each component and its latency contribution. It identifies bottlenecks, errors, and performance anomalies at the individual service/call level.
A is wrong: CloudWatch Metrics shows aggregate metrics (CPU, memory) but does not trace individual requests through a distributed call chain.
B is wrong: Log Insights can query structured logs but does not provide request-level distributed tracing or visualise service dependencies.
D is wrong: Amazon Inspector is a vulnerability assessment service, not a performance tracing tool.

38.4 Domain 4 — Migration & Modernisation

Question 7 SAP (migration strategy selection)

A company wants to migrate 200 servers from on-premises to AWS as quickly as possible with no application code changes. They have a 1 Gbps Direct Connect connection. Their largest server holds 50 TB of data on network-attached storage. Which combination of tools should they use? (Select TWO.)

A. Use AWS Application Migration Service (MGN) to replicate server operating systems and data continuously over Direct Connect, with a minimal-downtime cutover window.

B. Use AWS Snowball Edge to transport the 50 TB NAS dataset to AWS offline, then use DataSync for ongoing delta sync until cutover.

C. Use AWS DMS to migrate the 200 servers' data directly.

D. Use VM Import/Export for all 200 servers simultaneously.

E. Use S3 Transfer Acceleration for the 50 TB dataset.

Correct Answer: A, B

A is correct: AWS MGN (Application Migration Service) is the primary tool for lift-and-shift server migrations. It continuously replicates server disks to AWS (including over Direct Connect), then allows a low-downtime cutover. It supports hundreds of servers.
B is correct: transferring 50 TB over a 1 Gbps link at full utilisation takes ≈111 hours (4.6 days) — barely feasible but risky if the link is shared. Using Snowball Edge for the initial bulk load, then DataSync for delta synchronisation until cutover, is the standard AWS recommendation for large datasets.
C is wrong: DMS migrates databases, not entire server operating systems.
D is wrong: VM Import/Export is a one-off tool for individual VMs and does not scale well or provide continuous replication for 200 servers.
E is wrong: S3 Transfer Acceleration speeds up uploads to S3 (for objects), not for 50 TB NAS server data in the context of a server OS migration.

Question 8 SAP (modernisation / decoupling)

A monolithic application on EC2 processes customer orders synchronously. During peak sales, the EC2 fleet cannot scale fast enough and orders are lost. The company wants to modernise with *minimal code changes* and ensure no order is lost. What is the BEST first step?

A. Rewrite the entire application as Lambda functions with DynamoDB.

B. Place an SQS queue between the order intake tier and the processing tier; the intake layer writes order messages to SQS, and the processing tier reads from SQS at its own pace.

C. Enable Auto Scaling with a lower CPU target threshold so instances scale out faster.

D. Move the application to ECS Fargate containers for automatic scaling.

Correct Answer: B

B is correct: inserting an SQS queue decouples intake from processing with minimal code changes (just change the processing call to write to SQS). Orders are durably stored in the queue (up to 14 days) — none are lost. The processing tier drains the queue independently and can scale out without losing messages during scale-up lag.
A is wrong: a full rewrite to Lambda + DynamoDB is a Refactor strategy — high effort, not minimal code changes.
C is wrong: Auto Scaling has a launch delay (instance warm-up); during that window, orders can still be lost if the burst is immediate.
D is wrong: moving to Fargate still has the same synchronous coupling problem — processing still cannot keep up with instantaneous bursts.
Pattern: SQS as a buffer between tiers is the canonical answer for "no data loss during traffic spikes with minimal changes."

Question 9 SAP (cost optimisation)

A company runs 50 EC2 instances for a steady-state production workload (24/7) and 20 EC2 instances for batch jobs that run between 6 PM and 6 AM on weekdays. The company wants to minimise cost over a 3-year period. Which purchasing strategy should they use?

A. On-Demand for all instances.
B. 3-year Standard Reserved Instances for the 50 production instances; Spot Instances for the 20 batch job instances with an SQS queue to requeue interrupted jobs.
C. 1-year Convertible Reserved Instances for all 70 instances.
D. Savings Plans (Compute) for the 50 production instances; On-Demand for the 20 batch instances.

Correct Answer: B

B is correct: for 24/7 steady-state, 3-year Standard RIs provide the maximum discount (up to 72%). For batch jobs that are fault-tolerant (can be interrupted and requeued via SQS), Spot Instances provide up to 90% discount. This combination minimises cost for both workload types.
A is wrong: On-Demand for everything is the most expensive option.
C is wrong: 1-year Convertible RIs for batch instances is wasteful — the instances don't run 24/7 so the reserved capacity is unused for 12 hours per day.
D is partly right (Savings Plans for steady-state is valid) but On-Demand for batch is more expensive than Spot. Also, Savings Plans Compute may be better than RIs in some cases for flexibility, but the question asks for minimum cost — Spot for interruptible batch is the clear winner.
Key rule: Spot = fault-tolerant + interruptible. Standard RI = predictable 24/7. Savings Plans = flexible but lower max discount than Standard RIs.